

NEUROSCIENCE

A brain proteomic signature of incipient Alzheimer's disease in young *APOE* ϵ 4 carriers identifies novel drug targets

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Aptamer-based proteomics revealed differentially abundant proteins in Alzheimer's disease (AD) brains in the Baltimore Longitudinal Study of Aging and Religious Orders Study (mean age, 89 ± 9 years). A subset of these proteins was also differentially abundant in the brains of young *APOE* ϵ 4 carriers relative to noncarriers (mean age, 39 ± 6 years). Several of these proteins represent targets of approved and experimental drugs for other indications and were validated using orthogonal methods in independent human brain tissue samples as well as in transgenic AD models. Using cell culture-based phenotypic assays, we showed that drugs targeting the cytokine transducer STAT3 and the Src family tyrosine kinases, YES1 and FYN, rescued molecular phenotypes relevant to AD pathogenesis. Our findings may accelerate the development of effective interventions targeting the earliest molecular triggers of AD.

INTRODUCTION

The ϵ 4 allele of the apolipoprotein-E (*APOE*) gene is the most robust genetic risk factor for sporadic or late-onset Alzheimer's disease (AD). Heterozygous carriers of the ϵ 4 allele are at 3 to 4 times greater risk of AD, while homozygous individuals are at 10 times greater risk relative to noncarriers (1–3). *APOE* ϵ 4 carriers also have an earlier age at onset of AD (4). Although found more than two decades

ago (5), the precise mechanisms mediating *APOE* ϵ 4-associated risk of AD remain unclear, and the promise of *APOE*-based AD treatments remains unrealized.

APOE ϵ 4 carriers accumulate AD neuropathology early in adulthood. A neuroimaging meta-analysis showed that 15% of non-demented *APOE* ϵ 4 homozygous individuals showed evidence of cerebral amyloid accumulation at 40 years of age (6). In a recent postmortem autopsy study of young *APOE* ϵ 4 carriers, Pletnikova *et al.* (7) found that more than 40% of ϵ 4 heterozygous and 80% of ϵ 4 homozygous individuals between 40 and 49 years of age had diffuse brain amyloid plaques, compared to less than 1% of non-carriers. In addition, a study of regional cerebral glucose metabolism demonstrated that young, cognitively normal *APOE* ϵ 4 carriers between 20 and 39 years showed regional patterns of hypometabolism similar to those observed in patients with AD (8). Similarly, structural imaging studies have shown that among individuals under the age of 40, *APOE* ϵ 4 carriers have reduced cortical thickness (9), reduced gray matter volume, and worse cognitive performance (10).

This suggests that *APOE* ϵ 4 contributes to AD risk over the life course several decades before disease onset. The underlying biological pathways that connect *APOE* genotype with the development of pathology that eventually leads to AD, however, remain unknown. Identifying the precise biological mechanisms that operate in young *APOE* ϵ 4 carriers to accelerate AD pathogenesis is critical to both understanding *APOE*-related AD risk and the eventual development of *APOE*-guided AD treatments.

To characterize these biological pathways, we undertook a three-stage study (Fig. 1). We first identified proteins altered in the brains of AD individuals that are also dysregulated in young *APOE* ϵ 4 carriers and may therefore confer risk for future AD. We then validated those proteins across multiple independent cohorts including brain tissue samples from human studies and two transgenic AD mouse

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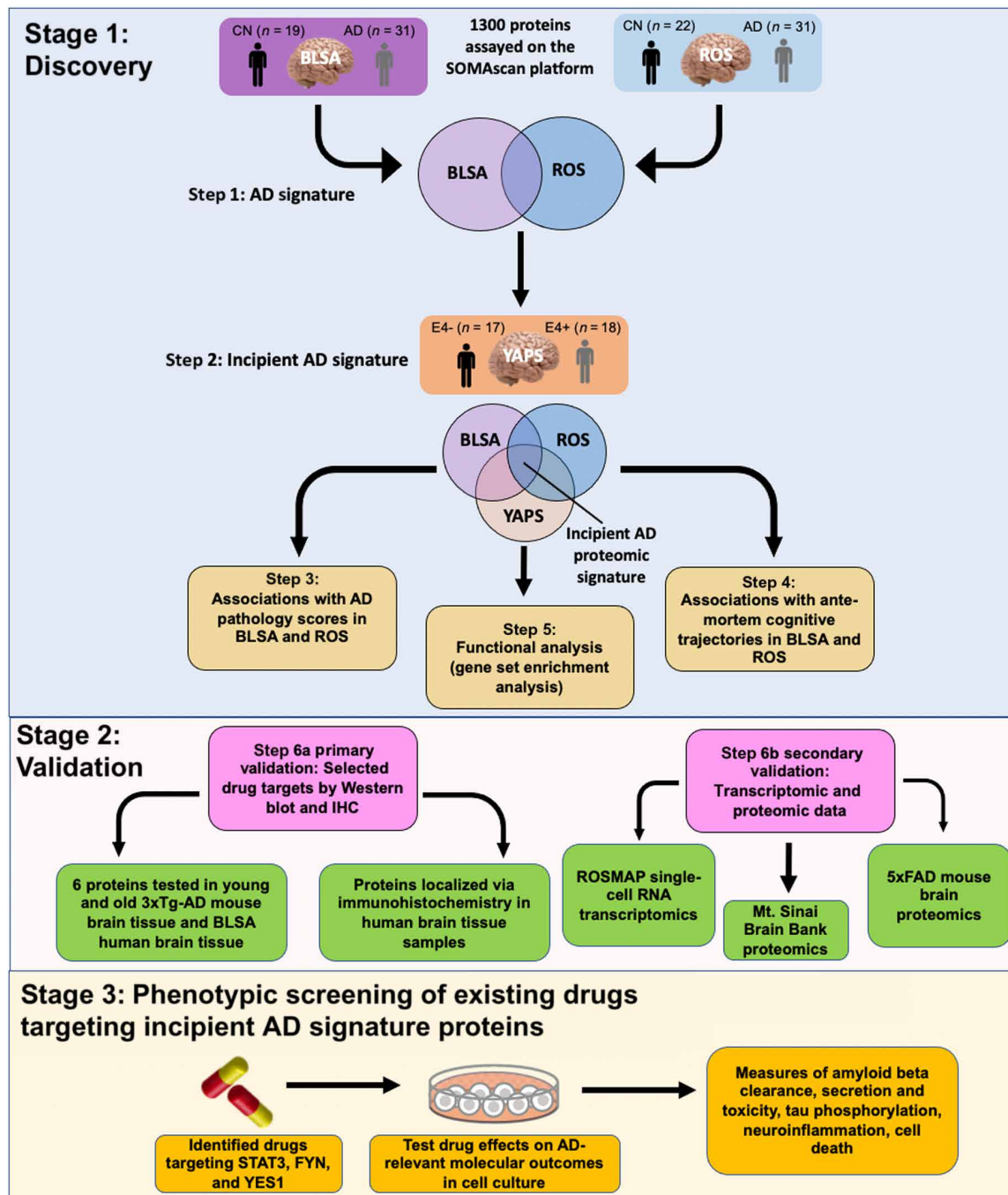


Fig. 1. Study design. Stage 1: In step 1, aptamer-based proteomics revealed protein-level differences in brain samples from both of two AD cohorts (BLSA and ROS; AD proteomic signature). In step 2, AD proteomic signature proteins were assessed in a cohort of young *APOE* $\epsilon 4$ carriers and noncarriers (YAPS). Proteins differing in all three cohorts were defined as an incipient AD proteomic signature. In steps 3 and 4, we tested associations between the incipient AD proteomic signature and both AD pathology and ante-mortem cognitive trajectories. In step 5, we compared GSEA in YAPS to the older adult samples to identify AD-related biologic pathways also altered in young *APOE* $\epsilon 4$ individuals. Stage 2: In step 6a, we validated a subset of proteins that are targets of approved and experimental drugs for non-AD indications, as biological pathways represented by these proteins may present plausible novel AD therapeutic targets. We assessed their levels in brain tissue using Western blot (WB) in the 3xTg-AD mouse model, as well as in a subset of AD and CN BLSA participants. We additionally assessed subcellular localization using immunohistochemistry (IHC) in brain samples from participants without AD pathology. In step 6b, we validated the incipient AD proteomic signature proteins in three publicly available datasets using orthogonal methods: mass spectrometry (MS)-based human brain proteomics (Mt. Sinai Brain Bank), MS-based mouse brain proteomics (5xFAD transgenic mouse AD model), and a single-cell human neuronal RNA transcriptomic dataset (ROSMAP). In stage 3: we performed phenotypic screening of existing drugs that are FDA-approved or in clinical trials for other indications targeting STAT3, YES1, and FYN in cell culture to test their ability to rescue AD-relevant phenotypes. BLSA, Baltimore Longitudinal Study of Aging; YAPS, Young *APOE* Postmortem Study; CN, cognitively normal; ROS, Religious Orders Study.

models using orthogonal proteomic and transcriptomic methods. Last, using phenotypic assays in cell culture models, we showed that drugs targeting three of these proteins—STAT3, YES1, and FYN—rescue distinct molecular phenotypes relevant to AD pathogenesis.

Stage 1: Discovery

We first identified an AD proteomic signature by comparing brain tissue protein levels in two independent, older adult postmortem samples of AD and age-matched cognitively normal controls (CN). We then evaluated proteins identified in the AD proteomic signature in a young, postmortem sample of *APOE* $\epsilon 4$ carriers (i.e., *APOE* $\epsilon 3/4$) and noncarriers to derive an incipient AD proteomic signature, i.e., the subset of differentially abundant proteins in both young *APOE* $\epsilon 4$ carriers relative to noncarriers and older adult AD individuals relative to CN (step 2). To assess the relationship between the incipient AD proteomic signature and endophenotypes of AD, we examined associations with severity of AD pathology at death [i.e., Consortium to Establish a Registry for Alzheimer's Disease (CERAD) and Braak scores] (step 3) and with antemortem trajectories of cognitive decline before the onset of AD (step 4). We additionally identified AD-related biologic pathways that may be altered in young *APOE* $\epsilon 4$ individuals by comparing results of gene set enrichment analyses (GSEA) in the young sample to the older adult samples (i.e., AD individuals relative to CN; step 5).

Stage 2: Validation

Step 6a: Primary validation

We selected a subset of six proteins from the incipient AD proteomic signature that are targets of both approved and experimental drugs for non-AD indications. Our rationale was that the biological pathways represented by these proteins may reflect plausible novel therapeutic targets in AD. We assessed their levels using Western blotting in brain tissue samples from the 3xTg-AD mouse model of AD as well as AD and CN human brain tissue samples. We additionally assessed the subcellular localization of these proteins in the human brain using immunohistochemistry.

Step 6b: Secondary validation

We additionally assessed protein and transcript levels of all 25 proteins comprising the incipient AD proteomic signature in multiple publicly available datasets using orthogonal methods, including mass spectrometry (MS)-based proteomics in an independent cohort of AD and CN human brain tissue samples, the 5xFAD transgenic mouse model of AD, and single-cell neuronal transcriptomic data from AD and CN individuals.

Stage 3: Phenotypic screening of drugs targeting the incipient AD signature

Among the subset of proteins selected in stage 2, we tested whether drugs targeting the cytokine transducer STAT3 and Src family tyrosine kinases YES1 and FYN could rescue molecular phenotypes relevant to AD pathogenesis. To accomplish this, we used cell culture-based phenotypic assays that provided readouts relevant to A β , tau pathology, neuroinflammation, and cell death.

RESULTS

Stage 1: Discovery of the incipient AD signature

Demographic characteristics

The characteristics of the three cohorts are summarized in Table 1. In the Baltimore Longitudinal Study of Aging (BLSA) sample, the

AD and CN groups did not differ significantly in age at death, sex, *APOE* $\epsilon 4$ carrier status, and postmortem interval (PMI). The AD group included a higher number of white participants (race) compared to CN samples. As expected, AD and CN groups varied significantly in Mini Mental State Exam (MMSE) score, severity of neuritic plaques (CERAD scores), and neurofibrillary tangles (Braak scores) with lower cognition and higher levels of pathology in the AD group.

In the Religious Orders Study (ROS) sample, the AD and CN groups did not vary significantly in race, *APOE* $\epsilon 4$ carrier status, and PMI. The AD group was significantly older at death and more likely female (sex) compared to CN. As expected, AD and CN groups varied significantly in MMSE score, severity of neuritic plaques (CERAD scores), and neurofibrillary tangles (Braak scores), with lower cognition and higher levels of pathology in the AD group.

Table 1 additionally summarizes differences across the BLSA and ROS cohorts. Considering the total sample, BLSA and ROS varied significantly in sex, race, and PMI. Comparing by group (e.g., BLSA AD/CN compared to ROS AD/CN), the BLSA AD group had an earlier age of onset, had a longer disease duration, was lower percentage female, had higher MMSE score, had higher Braak score, and had a longer PMI compared to the ROS AD group.

The Young *APOE* Postmortem Study (YAPS) cohort had a mean age of 39 years, was approximately 50% *APOE* $\epsilon 3/4$ by design, and contained 29% non-white individuals. There were no differences in other demographic characteristics between *APOE* $\epsilon 4+$ and *APOE* $\epsilon 4-$ groups in YAPS.

AD proteomic signature

To determine a brain proteomic signature of AD, we compared protein levels between AD and CN individuals in the ROS and BLSA cohorts in the inferior temporal gyrus (ITG) and middle frontal gyrus (MFG) brain regions. Of the 1300 proteins measured, in the BLSA, we identified 0 and 254 differentially abundant proteins [false discovery rate (FDR)-adjusted $P < 0.10$] in the ITG and MFG, respectively. In the ROS cohort, we identified 244 and 4 differentially abundant proteins (FDR-adjusted $P < 0.10$) in the ITG and MFG, respectively (244 unique proteins in total). The complete results of these analyses are included in table S2.

To establish an AD proteomic signature across both cohorts, we tested the intersection between differentially abundant proteins in the BLSA and ROS cohorts. This resulted in 120 unique proteins that were differentially abundant in both BLSA and ROS in either the ITG or MFG (Fig. 2A). Of these 120 proteins, 84 were lower in AD compared to CN and 34 were higher in AD. Only two displayed inconsistent direction between ROS and BLSA (i.e., higher in one cohort and lower in the other).

Incipient AD proteomic signature

Our primary motivation was to determine proteomic alterations in young *APOE* $\epsilon 4$ carriers that may drive risk for subsequent AD. We therefore determined which of the 120 proteins altered in both AD cohorts were also dysregulated between young *APOE* $\epsilon 4$ carriers and noncarriers in the YAPS cohort. Of the 120 proteins, 16 and 14 proteins were significantly different ($P < 0.05$) between *APOE* $\epsilon 4$ carriers and noncarriers in the ITG and MFG, respectively, resulting in 25 unique proteins defining the incipient AD proteomic signature (Fig. 2A). Of these proteins, 24 were increased in young *APOE* $\epsilon 4$ carriers, while 1 protein was reduced relative to *APOE* $\epsilon 4$ noncarriers (Fig. 2B). For proteins significantly altered in both ITG and MFG brain regions, direction was always consistent in the

Table 1. Cohort demographics. BLSA, Baltimore Longitudinal Study of Aging; YAPS, Young APOE Postmortem Study; ROS, Religious Orders Study; Disease duration, age death – age onset; MMSE, Mini Mental State Exam (last available before death); APOE ε4+, APOE ε4 carrier; APOE ε4–, APOE ε4 noncarrier. Note: The Wilcoxon rank sum (Mann-Whitney) test and χ^2 test were used to test for differences in continuous and binary variables, respectively.

BLSA			
	Total sample	AD	CN
	N = 52	N = 31	N = 21
Age at death, mean (SD)	86.26 (10.50)*	88.46 (8.30)*	83.02 (12.63)
Age of onset, mean (SD)	–	78.81 (9.92)*	
Disease duration, mean (SD)	–	9.65 (4.47)*	
Sex, n (% female)	22 (42.31)*	15 (48.39)*	7 (33.33)
Race, n (% white)	49 (94.23)	31 (100)†	18 (85.71)†
APOE ε4 carrier, n (%)	14 (28.00)	9 (31.03)	5 (23.81)
MMSE, mean (SD)	24.66 (5.59)	21.60 (6.05)*†	28.06 (1.98)†
CERAD, mean (SD)	1.73 (1.34)	2.77 (0.43)†	0.19 (0.40)†
Braak, mean (SD)	4.10 (1.67)	5.13 (1.12)*†	2.57 (1.08)†
PMI (hours), mean (SD)	15.58 (10.01)*	16.25 (11.73)*	14.53 (6.63)*
ROS			
	Total sample	AD	CN
	N = 53	N = 31	N = 22
Age at death, mean (SD)	90.58 (6.53)*	92.81 (5.45)*†	87.44 (6.75)†
Age of onset, mean (SD)	–	89.14 (5.71)*	–
Disease duration, mean (SD)	–	3.67 (2.94)*	–
Sex, n (% female)	40 (75.47)*	27 (87.10)*†	13 (59.09)†
Race, n (% white)	53 (100.00)	31 (100.00)	22 (100.00)
APOE ε4 carrier, n (%)	13 (25.49)	9 (31.03)	4 (18.18)
MMSE, mean (SD)	19.95 (10.10)	12.77 (8.43)*†	28.50 (1.59)†
CERAD, mean (SD)	1.64 (1.26)	2.58 (0.50)†	0.32 (0.65)†
Braak, mean (SD)	3.89 (1.33)	4.65 (0.66)*†	2.82 (1.30)†
PMI (hours), mean (SD)	9.05 (5.32)*	8.67 (5.36)*	9.59 (5.35)*
YAPS			
	Total sample	APOE ε4+	APOE ε4–
	N = 35	N = 18	N = 17
Age at death, mean (SD)	38.97 (5.95)	39.06 (6.15)	38.88 (5.93)
Sex, n (% female)	13 (37.14)	6 (33.33)	7 (41.18)
Race, n (% white)	25 (71.43)	14 (77.78)	11 (64.71)
PMI (hours), mean (SD)	23.14 (11.42)	24.39 (12.46)	21.82 (10.42)

*P < 0.05 comparing BLSA to ROS (e.g., AD in BLSA compared to AD in ROS). †P < 0.05 comparing AD to CN.

YAPS cohort. Nearly all proteins in the incipient AD proteomic signature displayed an opposite direction of association in YAPS compared to BLSA and ROS such that protein levels were increased in young APOE ε4 carriers relative to noncarriers and reduced in AD relative to controls. Of the 25 proteins, only one protein (TBP) displayed an inconsistent direction of association between BLSA and ROS (increased in YAPS and ROS but decreased in BLSA; indicated by an asterisk in Fig. 2D).

We then proceeded to test the statistical significance of the overlap between these three proteomic signatures by implementing multiset

intersection analysis through the SuperExactTest developed by Wang *et al.* (11). This procedure computes the statistical distributions of multiset intersections using combinatorial theory and efficiently calculates their exact probabilities. The YAPS, BLSA, and ROS proteomic signatures, including 86, 254, and 244 significantly altered proteins, respectively, share 25 proteins, and this intersection, defined as the incipient AD proteomic signature, is highly significant (SuperExactTest, $P < 0.00001$). This analysis indicates that the observed signature of 25 proteins in the incipient AD proteomic signature significantly exceeds the expected, or null, signature of 3.15 proteins (i.e., 7.9-fold

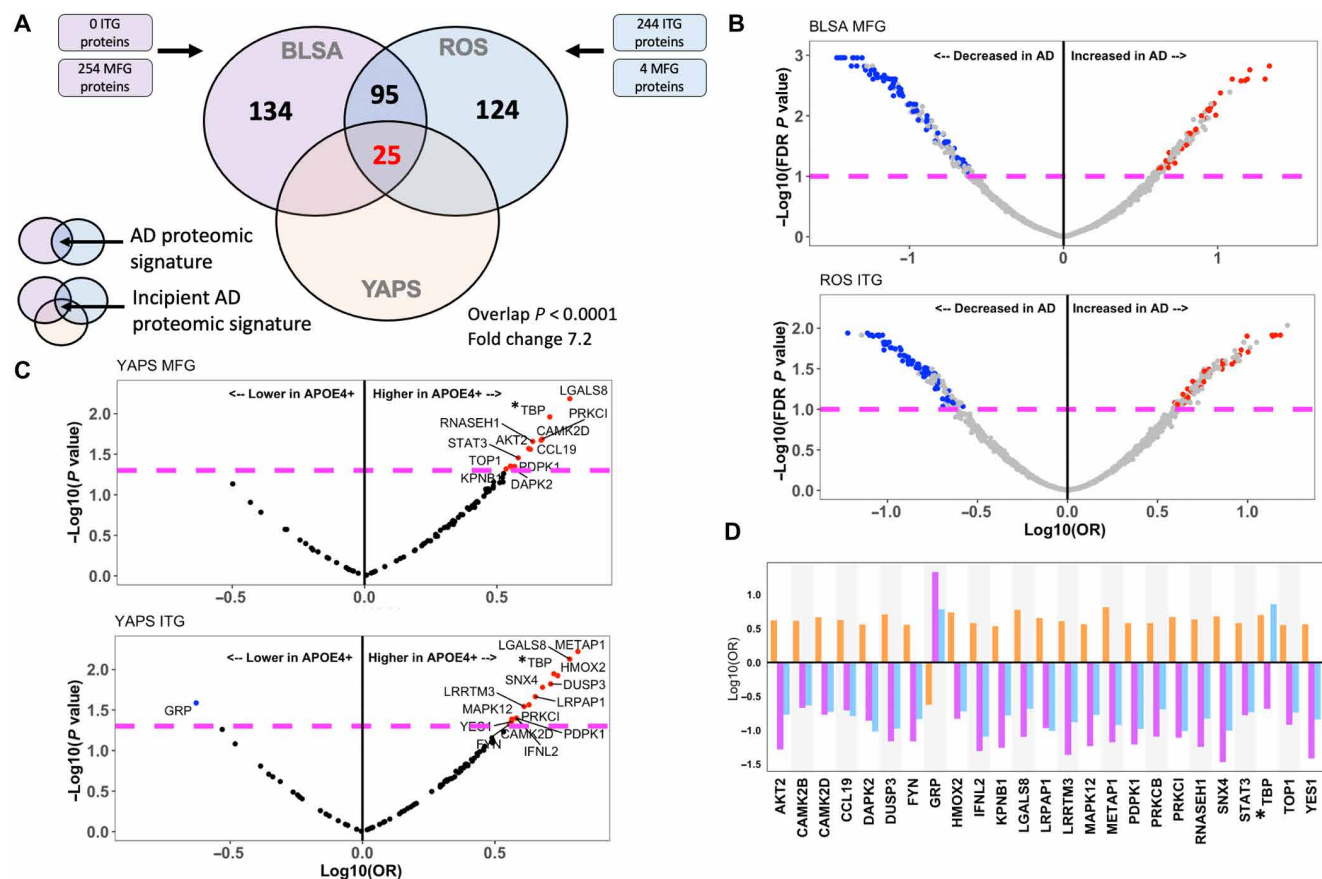


Fig. 2. Identifying an incipient AD proteomic signature. (A) Proportional odds models to identify an incipient AD proteomic signature. Comparing AD and CN individuals in BLSA and ROS, we identified an AD proteomic signature as 120 unique proteins across the ITG and MFG ($\text{FDR } P < 0.10$). Two-way intersection significance was determined using SuperExactTest ($P < 0.00001$). We then analyzed these 120 proteins in YAPS to establish an incipient AD proteomic signature (25 proteins in red) as the overlap across all three cohorts. Three-way intersection significance was determined using SuperExactTest ($P < 0.00001$). (B) Proteins in BLSA and ROS defining the 120 protein AD proteomic signature. Pink dashes indicate $\text{FDR } P = 0.10$. The black vertical line represents no difference in protein concentration between AD and CN. A total of 1300 proteins are presented; gray points indicate proteins that do not meet the FDR threshold. Blue and red points indicate proteins significantly lower and higher, respectively, in AD individuals in both cohorts. (C) AD proteomic signature proteins overlapping with proteins differentially abundant in YAPS (the 25-protein incipient AD signature). Pink dashes represent $P = 0.05$. The 120-protein AD proteomic signature is the background; gray points indicate proteins that do not meet the P value threshold. Blue and red points indicate proteins significantly lower and higher, respectively, in $\text{APOE } \epsilon 4+$ individuals. (D) Protein levels of the 25-protein incipient AD proteomic signature (region with lower P value visualized). Y axis indicates $\log_{10}(\text{OR})$, with positive values indicating increased protein levels in $\text{APOE } \epsilon 4+$ (YAPS) or AD (BLSA and ROS) and negative values indicating lower levels. OR, odds ratio. * indicates opposite direction of abundance in AD/CN between BLSA and ROS.

enrichment) had the three cohort-specific signatures been selected at random from the original list of 1300 proteins. In addition, the overlap between ROS and BLSA proteomic signatures, defined as the AD proteomic signature, consisting of 120 significant proteins, is similarly highly significant (SuperExactTest, $P < 0.00001$) and exceeds the expected overlap of 47.6 proteins (i.e., 2.52-fold enrichment) (table S3).

To determine whether proteins in the incipient AD proteomic signature were functionally related, we performed pathway enrichment analysis using the 25-protein list. This analysis identified several significantly enriched pathways among the signature, including cytokine signaling, tyrosine kinase signaling, cell migration, and other signaling pathways (table S4). We additionally examined the protein-protein interactions in the signature using the StringDB database (<https://string-db.org>). This identified numerous significant interactions, including identification of STAT3, FYN, and YES1 as central nodes (fig. S1). An extensive annotation of the biological

roles and clinical significance of the incipient AD proteomic signature is included in table S5.

Sensitivity analysis

To verify that the differential abundance of proteins in the incipient AD proteomic signature was not driven by the presence of $\text{APOE } \epsilon 4+$ AD individuals, we excluded $\text{APOE } \epsilon 4+$ individuals from the BLSA and ROS cohorts. Results were generally consistent with the main analysis: 24 of 25 proteins from the incipient AD proteomic signature remained statistically significant (table S6).

Similarly, we restricted the sample to only $\text{APOE } \epsilon 4+$ individuals in ROS and BLSA to perform similar comparisons between AD and CN. Despite the small sample size in this comparison (AD/CN, $N = 9/4$ in BLSA and $9/4$ in ROS), seven proteins from the incipient AD proteomic signature remained statistically significant, and the direction of association for all 25 proteins remained consistent with the main analysis (table S7).

The incipient AD proteomic signature is associated with severity of AD pathology: Braak and CERAD scores

In the ITG and MFG, 11 of 25 and 15 of 25 proteins, respectively, from the incipient AD proteomic signature were also significantly ($P < 0.05$) associated with severity of neurofibrillary pathology (Braak scores), with consistent direction in both the BLSA and ROS cohorts (Fig. 3A). In the ITG and MFG, 11 of 25 and 21 of 25 proteins, respectively, from the incipient AD proteomic signature were significantly ($P < 0.05$) associated with severity of neuritic plaque burden (CERAD scores), with consistent direction in both the ROS and BLSA cohorts (Fig. 3B).

With the exception of gastrin-releasing peptide (GRP), which was associated with higher CERAD scores in the MFG of both cohorts, all significant shared associations in the BLSA and ROS cohorts indicated that a lower protein level was associated with a higher pathology score. In both BLSA and ROS, seven proteins (CCL19, METAP1, DUSP3, SNX4, IFNL2, YES1, and PDPK1) displayed significant associations with both CERAD and Braak scores in both the ITG and MFG brain regions.

The incipient AD proteomic signature is associated with antemortem trajectories of cognitive performance before development of AD

In either the ITG or MFG, 20 of 25 proteins in the incipient AD proteomic signature were significantly ($P < 0.05$) associated with longitudinal trajectories of MMSE scores in AD individuals from

BLSA or ROS (Fig. 3C). Six proteins were significantly associated with longitudinal trajectories of MMSE scores, with a consistent direction in both BLSA and ROS: CAMK2B, CAMK2D, and LRRTM3 were associated with a slower decline in MMSE, while DUSP3, KPNB1, and LRPAP1 were associated with a faster decline in MMSE in both cohorts.

GSEA identifies pathways dysregulated in both young APOE $\epsilon 4$ carriers and AD

To derive a global understanding of molecular pathways and biological functions implicated by the proteomic alterations associated with APOE $\epsilon 4$ carrier status in young individuals, we performed GSEA on all proteins included in our dataset. All 1300 measured proteins were utilized in YAPS, BLSA, and ROS so that analyses were not biased toward only highly significant proteins but rather provided a comprehensive overview of proteomic alterations in each cohort. In the YAPS cohort, this analysis identified 44 significantly enriched gene sets (FDR-adjusted $P < 0.05$) in either the ITG or MFG. The majority (41 of 44) of these gene sets were overexpressed [i.e., positive normalized enrichment score (NES)] in the YAPS cohort, indicating an up-regulation of these pathways and their biological functions in young APOE $\epsilon 4$ carriers relative to noncarriers. These results signal an alteration in several biological functions including calcium signaling, synaptic transmission, insulin signaling, immune response, and cell-cell adhesion.

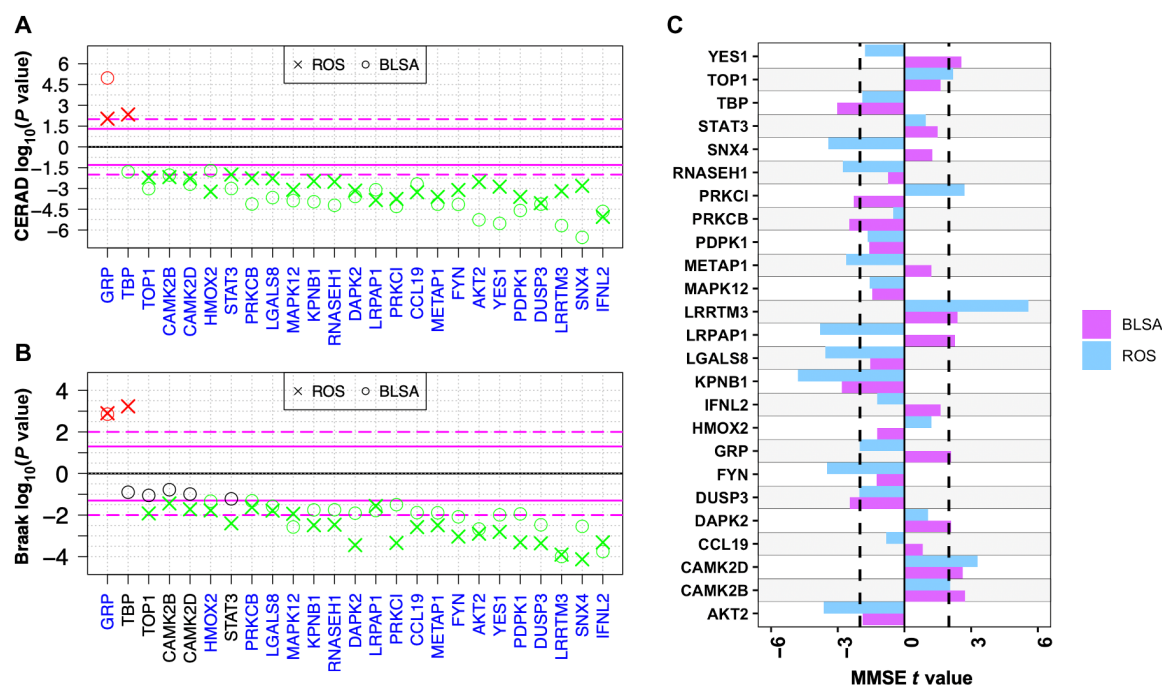


Fig. 3. Associations between the incipient AD proteomic signature with severity of AD pathology and longitudinal trajectories of antemortem cognitive performance.

Partial correlation analyses between 25 proteins comprising the incipient AD proteomic signature and (A) Braak and (B) CERAD scores in the BLSA and ROS cohorts in either the ITG or MFG (brain region with lower P value visualized). The y axis indicates the $\log_{10}(P$ value), and protein names are indicated on the x axis. Positive values indicate that a higher protein concentration was associated with a higher pathology score, while negative values indicate that a higher protein concentration was associated with a lower pathology score. The solid pink line indicates $P = 0.05$; the dashed pink line indicates $P = 0.01$. Positive significant values (red) and negative significant values (green) indicate that higher protein concentration is associated with higher or lower neurofibrillary tangle pathology (Braak scores)/neuritic plaque burden (CERAD scores), respectively. Nonsignificant associations are in black. Blue protein names indicate a significant association in both ROS and BLSA. (C) Associations between the incipient AD proteomic signature and longitudinal trajectories in MMSE scores in AD individuals in the BLSA and ROS in the ITG or MFG (brain region with lower P value visualized). A negative t value indicates a negative association between the protein slope in MMSE scores over time (i.e., a higher protein level is associated with a faster/increased decline in MMSE). A positive t value indicates a positive association between the protein and slope in MMSE scores over time. Values beyond the dashed lines indicate $P < 0.05$.

We further sought to identify whether any of the gene sets enriched in the YAPS cohort were similarly enriched in either of the AD cohorts, i.e., BLSA or ROS. We therefore performed GSEA with the same gene sets analyzed in the YAPS cohort for BLSA and ROS separately (Fig. 4). This analysis showed that 17 of the 44 significantly enriched gene sets in YAPS were also enriched in either BLSA or ROS individuals (FDR-adjusted $P < 0.05$). Furthermore, we tested the significance of this overlap by performing Fisher's exact ratio tests between the number of overlaps at different FDR thresholds (0.05, 0.1, and 0.25), which gave Fisher test $P < 4.8 \times 10^{-12}$, indicating a statistically significant overlap in enriched gene sets between the YAPS and BLSA/ROS cohorts independent of the chosen FDR threshold. Full GSEA results are included in table S8.

Stage 2: Validation of the incipient AD signature

Primary validation: Western blot and immunohistochemical localization

Western blot analyses in 3xTg-AD brain homogenates. We first validated a subset of 6 proteins of the 25 proteins in the incipient AD proteomic signature using Western blotting in brain homogenates from the 3xTg-AD transgenic mouse model of AD. This model expresses mutant forms of human amyloid precursor protein (APP), presenilin-1, and tau, developing age-dependent accumulation of extracellular A β plaques, intracellular tau accumulation, oxidative stress, and cognitive deficits (12). We chose these proteins based on previous evidence suggesting that they are targeted by approved and experimental drugs for other diseases (table S9) and may therefore represent plausible therapeutic targets in AD through repurposing of existing drugs. Of the six proteins chosen, three showed significantly altered levels in the brains of transgenic 3xTg-AD mice relative to wild type (WT; $P < 0.05$) (Fig. 5A). In young transgenic mice aged 38 to 40 weeks, protein levels of DUSP3 and STAT3 were higher in transgenic mice relative to WT, whereas in older mice aged 83 to 136 weeks, protein levels of DUSP3 and TOP1 were lower in transgenic mice relative to WT. Full results are included in table S10, and Western blot images are included in fig. S2.

Western blot analyses in BLSA human brain homogenates. We next validated the same set of six proteins from the incipient AD proteomic signature using Western blotting in brain homogenates from a subset of AD and CN MFG brain tissue samples from the

BLSA (see table S10 for subject characteristics). Of the six proteins, two showed significantly altered levels in AD brain tissue compared to CN ($P < 0.05$; Fig. 5). FYN levels were reduced in AD, while LGALS8 levels were higher in AD relative to controls. STAT3 levels (marginally significant) were higher in AD relative to CN ($P = 0.053$) (Fig. 5B). Full results are included in table S10, and Western blot images are included in fig. S3.

Subcellular localization of proteins from the incipient AD signature in human brain. In these analyses, we attempted to establish the subcellular localization of selected proteins from the incipient AD signature in the human brain. Our goal in these experiments was to derive additional insights into plausible physiological roles of these proteins. We examined whether the subset of six proteins assayed by Western blotting in step 6a (Fig. 5; i.e., primary validation) could be detected in cortex within the nondiseased human brain using immunohistochemical colocalization of the proteins within distinct subcellular compartments. We used formalin-fixed paraffin-embedded (FFPE) tissue samples from the inferior parietal cortex obtained from five autopsy brain samples. The autopsies were performed at the Office of the Chief Medical Examiner (OCME) of the State of Maryland in Baltimore, and brains were obtained as previously described (13). Sample demographics are included in table S11. The brains were confirmed free of AD pathology determined by standard neuropathologic criteria according to CERAD guidelines (14). We first screened several commercially available antibodies for positive immunoreactivity (at least two antibodies for each selected target protein; table S12). Two target proteins (i.e., DUSP3 and LGALS8) showed good signal-to-noise ratio (SNR) with their corresponding antibodies and were selected for immunohistochemical colocalization experiments using additional antibodies against specific subcellular compartments.

DUSP3 showed positive immunoreactivity with both PSD95 (excitatory postsynaptic marker) and gephyrin (inhibitory postsynaptic marker) but not with synaptophysin (presynaptic marker), indicating that the protein is likely located in postsynaptic processes in both excitatory and inhibitory cortical neurons (Fig. 6). LGALS8 also showed positive immunoreactivity with LAMP1 (a lysosomal marker) rather than MTCO1 (mitochondrial marker) and Rab5 (early endosomal marker), indicating that the protein is likely mainly located in neuronal lysosomes (Fig. 7).

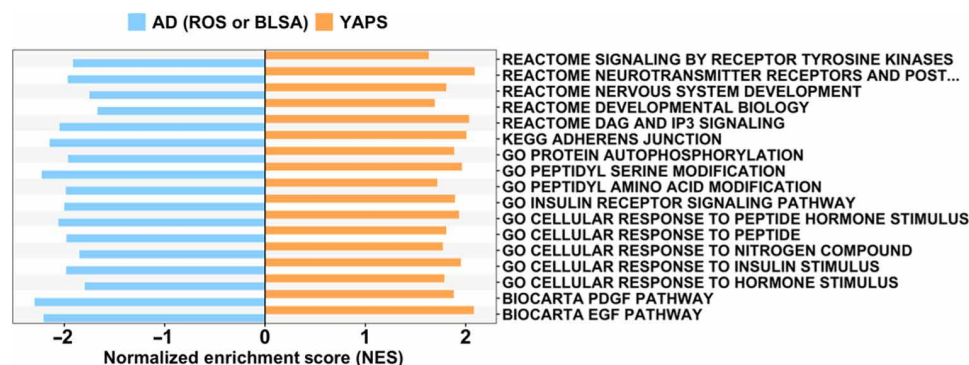


Fig. 4. GSEA identifies pathways dysregulated in both young *APOE* $\epsilon 4$ carriers and AD. Results of GSEA analyses for the 17 gene sets identified as dysregulated in both YAPS and at least one AD cohort. Comparing overlapping gene sets between YAPS and AD cohorts (ROS/BLSA), gene sets had significant and opposite expression in YAPS relative to the AD cohorts. The gene set name is given on the y axis, and NES is provided on the x axis. A positive NES indicates a significant positive enrichment of the gene set such that this pathway may be overexpressed in the cohort. A negative NES indicates a negative enrichment of the gene set such that this pathway may be underexpressed in the cohort. Bars are colored according to cohort.

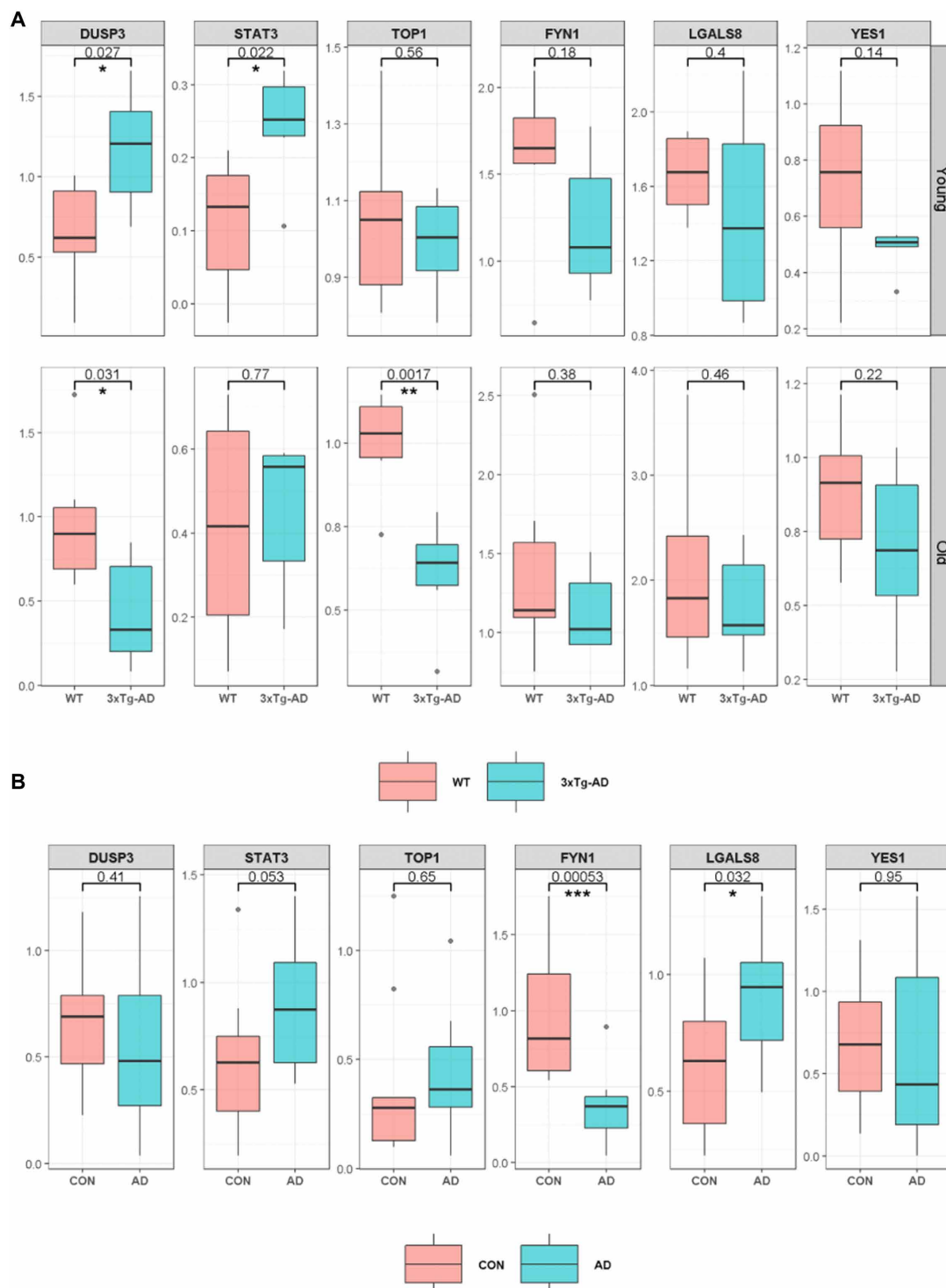


Fig. 5. Primary validation of incipient AD proteomic signature. Results of primary validation analyses performed using Western blotting in 3xTg-AD transgenic mice (AD) compared to wild type (WT) (A) and human brain tissue samples comparing AD to CN (B). The 3xTg-AD (n = 6) and WT (n = 6) mice included young (38 to 40 weeks) and old (83 to 136 weeks) mice. (A) Box plots represent differences in protein levels in brain homogenates between 3xTg-AD (blue) and WT (red) mice. The top and bottom rows indicate protein levels in young and old mice, respectively. (B) Box plots represent differences in protein levels in brain homogenates between AD (blue; n = 11) and CN (red; n = 11). Protein levels that were significantly different between 3xTg-AD and WT or AD and CN are indicated with * (P < 0.05) or ** (P < 0.01). 3xTg-AD, 3xTg-AD transgenic mouse model of AD; AD, Alzheimer's disease; WT, wild type; DUSP3, dual-specificity phosphatase 3; STAT3, signal transducer and activator of transcription 3; TOP1, DNA topoisomerase I; FYN1, FYN proto-oncogene tyrosine kinase; LGALS8, galectin-8; YES1, YES proto-oncogene tyrosine kinase.

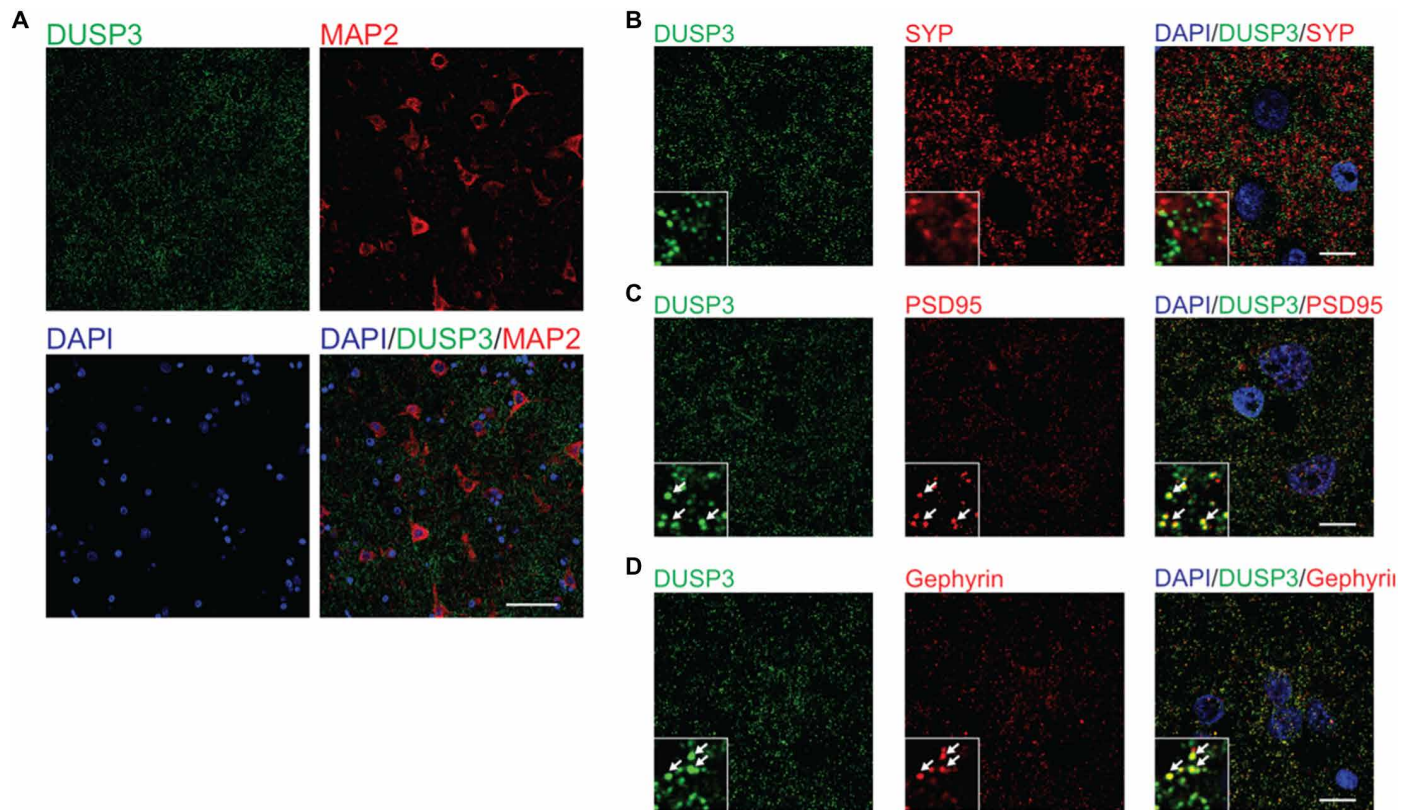


Fig. 6. DUSP3 immunocolocalization with subcellular markers. Representative images of DUSP3 antibody costained with various subcellular markers in neurons within the inferior parietal cortex. (A) Low-magnification image of DUSP3 and MAP2 (neuronal marker), (B) DUSP3 and SYP (synaptophysin, presynaptic marker), (C) DUSP3 and PSD95 (excitatory postsynaptic marker), and (D) DUSP3 and gephyrin (inhibitory postsynaptic marker). White arrows indicate the examples of colocalized immunoreactivity between antibodies. Scale bars, 50 μm (A) and 10 μm (B to D). Images (B) to (D) were captured at $\times 100$ magnification, with inset images cropped from the larger image.

Secondary validation: External validation of the incipient AD proteomic signature in independent datasets

In step 6b (secondary validation), we assessed all 25 proteins included in the incipient AD proteomic signature using orthogonal methods in independent samples, i.e., MS-based proteomics in (i) AD and CN samples from the Mount Sinai Brain Bank, (ii) 5xFAD transgenic mouse model of AD, and (iii) single-cell transcriptomics through RNA sequencing (scRNA-seq) from brain samples in the ROS and Memory and Aging Project (ROSMAP). The results of these validation analyses are summarized in Fig. 8, and complete results are in table S13. All 25 proteins validated in at least one of the three independent datasets, and 15 of 25 proteins validated in at least two. Primary data used for validation from these three sources were recently published (15, 16) and are publicly available.

Mount Sinai Brain Bank. In MS-based proteomic data in brain samples from the temporal cortex (Brodmann area 36) collected through the Mount Sinai Brain Bank, 21 of the 25 proteins in the incipient AD signature were quantified. We found that 12 of these proteins were differentially abundant in AD versus CN samples ($P < 0.05$).

5xFAD mouse model. 5xFAD mice express human APP and PSEN1 transgenes with five AD-linked mutations: the Swedish (K670N/M671L), Florida (I716V), and London (V717I) mutations in APP and the M146L and L286V mutations in PSEN1 (17). In MS-based proteomic data in brain cortical samples from the 5xFAD transgenic mouse model of AD, 21 of the 25 proteins in the incipient AD

signature were quantified. We found that six of these proteins were differentially abundant in transgenic mice versus WT at 6 months of age ($P < 0.05$).

ROSMAP scRNA-seq. In single-cell transcriptomic data from inhibitory and excitatory neurons from the dorsolateral prefrontal cortex samples in the ROSMAP cohort, 22 and 23 of 25 proteins in the incipient AD signature were quantified, respectively. We found that 19 (inhibitory neuron cell type) and 19 (excitatory neuron cell type) were differentially expressed in AD versus CN samples ($P < 0.05$).

Stage 3: Phenotypic screening of drugs targeting proteins in the incipient AD signature

To test whether drugs targeting proteins in the incipient AD signature may be plausible AD treatments, we used cell culture-based phenotypic assays to assess their ability to affect molecular outcomes relevant to AD. We nominated STAT3, YES1, and FYN as potential AD drug targets and approved/experimental drugs targeting them as candidate AD treatments. Drugs targeting STAT3 included in these studies were crizotinib [U.S. Food and Drug Administration (FDA)-approved for non-small cell lung cancer] (18), napabucasin (FDA-designated orphan drug status for colorectal and gastroesophageal cancer) (19), and C188-9 (currently in a phase 1 clinical trial of several cancers including lung, hepatocellular, and colorectal cancer; ClinicalTrials.gov identifier: NCT03195699). We selected dasatinib (FDA-approved for chronic myeloid leukemia) as a

LGALS8

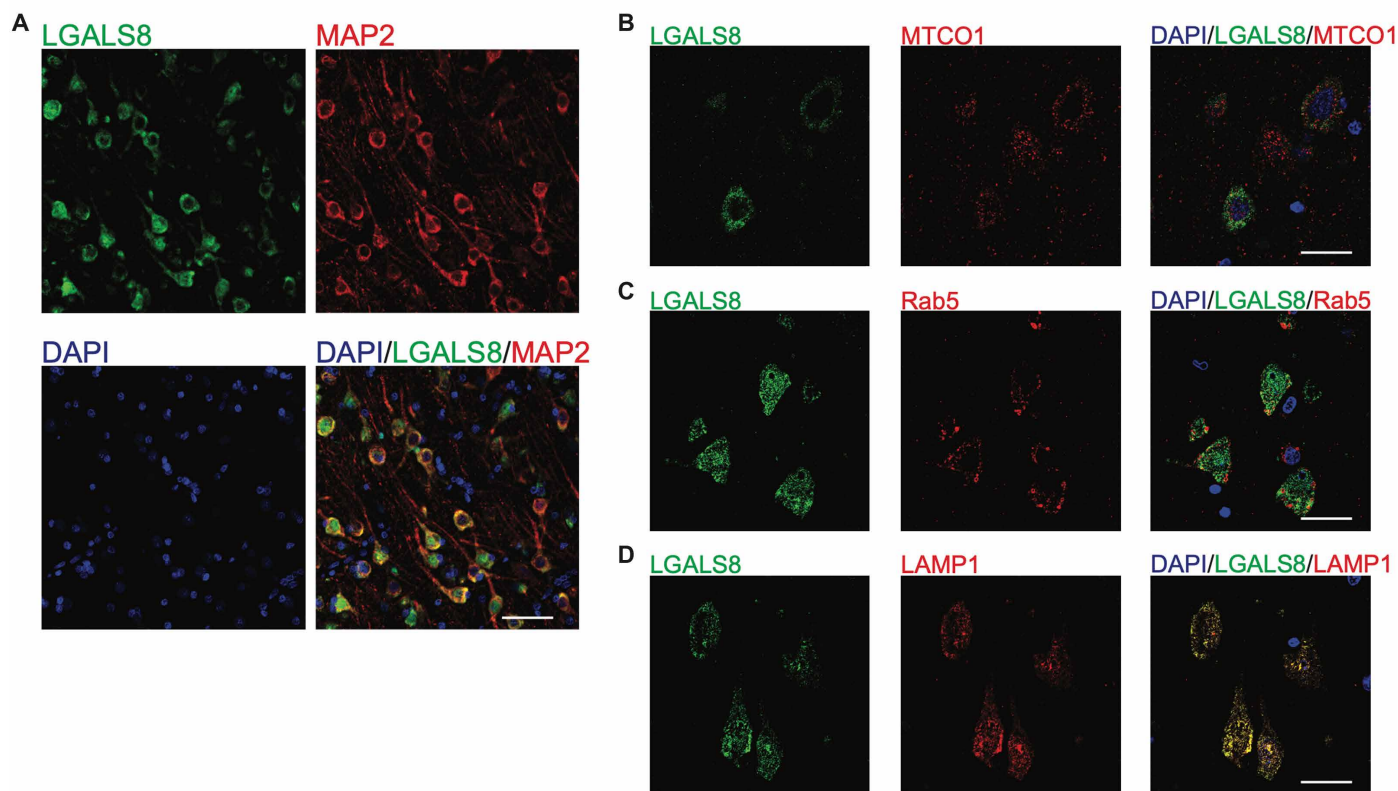


Fig. 7. LGALS8 immunocolocalization with subcellular markers. Representative images of LGALS8 antibody costained with various subcellular markers in inferior parietal cortex. (A) Low-magnification image of LGALS8 and MAP2 (neuronal marker), (B) LGALS8 and MTCO1 (mitochondrial marker), (C) LGALS8 and Rab5 (early endosomal marker), and (D) LGALS8 and LAMP1 (lysosomal marker). Scale bars, 50 μm (A) and 20 μm (B to D).

candidate AD treatment targeting the Src family tyrosine kinases YES1 and FYN (20).

As indicated in Fig. 9A, the STAT3 inhibitor C188-9 rescued three AD phenotypes: lipopolysaccharide (LPS)-induced neuroinflammation, tau phosphorylation, and A β secretion.

For the bacterial LPS-induced neuroinflammation assay, C188-9 significantly reduced the release of proinflammatory cytokines interleukin-6 (IL-6) relative to LPS (blue bar) at the highest concentration (0.6 μM) and significantly reduced the release of IL-1 β relative to LPS at both 0.3 and 0.6 μM concentration in BV2 microglial cells. For both cytokines, there was a significant increase relative to LPS at the lowest concentration (0.1 μM). There were no adverse effects on cell viability, and comparisons between LPS and the vehicular control (VC) as well as LPS and reference item dexamethasone (10 μM) (RI dexta) indicate that LPS stimulation successfully generated proinflammatory cytokines (e.g., IL-6 and IL-1 β) (results not shown). C188-9 also significantly reduced levels of phosphorylated tau231 (ptau) relative to the VC (blue bar) at the highest concentration (10 μM). Last, C188-9 significantly reduced levels of endogenous A β 1–42 and the A β 42:A β 40 ratio relative to the VC (blue bar) at the middle concentration (1 μM), with no adverse effects on cell viability (results not shown) in human APP-overexpressing H4 neuroglioma cells.

As indicated in Fig. 9B, the YES1/FYN inhibitor dasatinib rescued one AD phenotype: tau phosphorylation. Dasatinib significantly

reduced levels of total tau and ptau relative to the VC (blue bar) at all three concentrations (0.01, 0.1, and 1 μM) in the mutant tau441-overexpressing neuroblastoma cell line. Napabucasin and crizotinib did not rescue any of the phenotypes assayed. Full results from the phenotypic screening assays are included in table S14.

DISCUSSION

In this study, we established an incipient AD proteomic signature in young *APOE* $\epsilon 4$ carriers to characterize biological alterations in the brain that may precede AD onset by up to three decades. We accomplished this by using aptamer-based proteomics to first identify a brain proteomic signature of AD in two independent, well-characterized older adult cohorts and then by determining that several proteins in this signature were also altered in young *APOE* $\epsilon 4$ individuals without substantial AD pathology. This incipient AD proteomic signature consisted of 25 proteins altered across all three cohorts (BLSA, ROS, and YAPS). We first validated a subset of these proteins that are targeted by drugs used for non-AD indications using Western blotting in the 3xTg-AD mouse model of AD as well as in AD and CN samples from BLSA. In addition, we also confirmed the subcellular localization of two of these proteins within neuronal lysosomes and postsynaptic processes in excitatory and inhibitory neurons. We then validated this signature in three independent publicly available

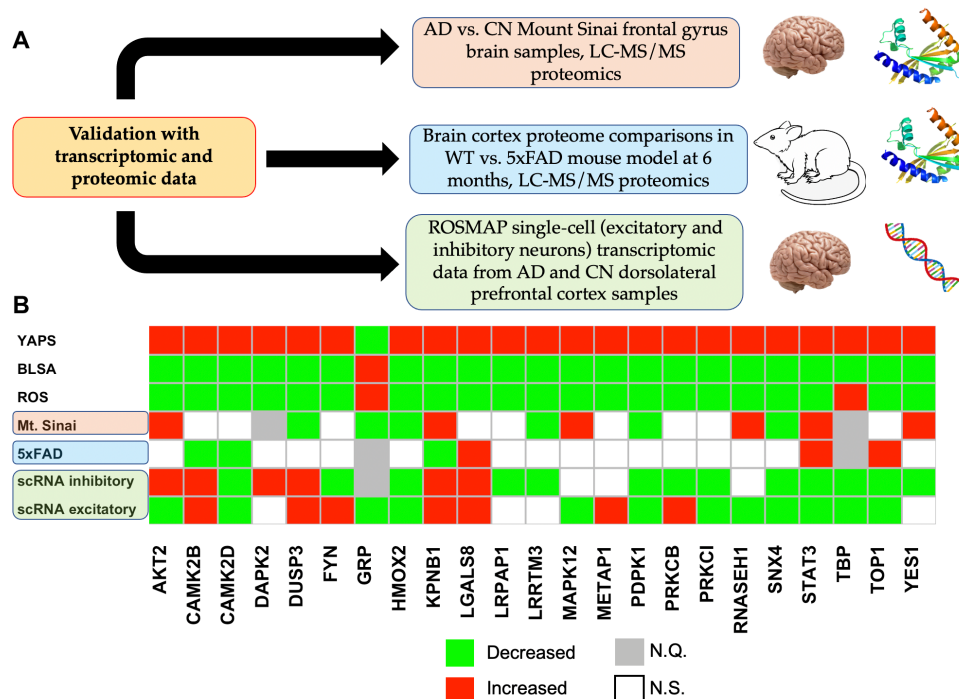


Fig. 8. Secondary validation of incipient AD proteomic signature. (A and B) Results of validation analyses carried out with publicly available proteomic and transcriptomic datasets. Colored squares represent proteins/genes that were differentially abundant/expressed between AD and CN or 5xFAD versus WT mice. Gray squares indicate proteins/transcripts that were not quantified (N.Q.). White squares indicate proteins/transcripts that were not significantly (N.S.) differentially abundant/expressed between AD and CN or 5xFAD versus WT mice. Significance was defined as $P < 0.05$. scRNA, single-cell RNA; LC-MS/MS, liquid chromatography with tandem MS.

datasets using orthogonal methods including MS-based proteomics in AD and CN samples as well as in the 5xFAD mouse model of AD. We additionally confirmed differential neuronal expression of several gene transcripts encoding these proteins in AD.

By identifying molecular correlates of *APOE* $\epsilon 4$ associated AD risk in young individuals, our results provide a window into very early biological perturbations occurring during the long preclinical phase of AD that may present novel therapeutic targets for disease modification. We provide evidence supporting this hypothesis by demonstrating that drugs targeting STAT3, YES1, and FYN, three proteins in the incipient AD proteomic signature, reduce neuroinflammation and tau phosphorylation as well as endogenous production of A β 42 in cell culture-based phenotypic assays. Together with recent findings that *APOE*-targeted immunotherapy reduces brain amyloid deposition and rescues cerebrovascular dysfunction in the 5xFAD mouse model (21), our results suggest that *APOE*-associated dysregulation in molecular pathways may offer a promising source of novel drug targets in AD.

Of the 25 proteins altered in young *APOE* $\epsilon 4$ carriers and in AD, some have established roles in both A β accumulation and tau phosphorylation—molecular events critical to the development of the two primary pathological hallmarks of AD. These include roles both in the amyloidogenic processing of the APP and in clearance of A β from the brain. For example, leucine-rich repeat transmembrane neuronal 3 (LRRTM3) is a synaptogenic adhesion molecule involved in synaptic assembly and promotes APP processing by β -secretase 1 (BACE1) (22–24), while sorting nexin 4 (SNX4) prevents BACE1 trafficking to lysosomes for degradation, thereby facilitating A β production (25, 26). On the other hand, LDL

receptor-related protein associated protein 1 (LRPAP1) facilitates LRP-mediated A β clearance across the blood-brain barrier (27, 28). It is interesting to note that polymorphic variation in both LRRTM3 and LRPAP1 has been associated with increased risk of late-onset AD, further suggesting potential causative roles for these proteins in AD pathogenesis (29, 30). We also observed several kinases that participate in multiple signaling cascades relevant to AD in the incipient AD proteomic signature. These include the atypical protein kinase C (aPKC) PKC- ι , mitogen-activated protein kinase 12 (MAPK12), a member of the p38 MAPK family, the Src family tyrosine kinases FYN and YES1 (31), and Ca²⁺/calmodulin (CaM)-dependent protein kinase II (CaMKII), the major postsynaptic protein at excitatory synapses (32). PKC- ι has been shown to mediate an increase in BACE activity, A β production, and tau phosphorylation and is known to be modulated by brain insulin levels (33). The p38 MAPKs phosphorylate microtubule-associated tau in addition to a broad range of proteins and have been shown to be important mediators of the senescence-associated secretory phenotype (SASP)—a chronic proinflammatory state in senescent cells, characterized by the secretion of numerous cytokines and chemokines (34). Previous studies from postmortem human brains have also reported activation of p38 MAPKs in early stages of AD (35). Altered protein levels of FYN, an important regulator of pathological tau aggregation and transducer of A β signaling, suggest that convergent dysregulation of both A β - and tau-related pathways may be an early feature of AD progression (36). The presence of two distinct isoforms of CaMKII in the incipient AD signature is especially interesting given its important roles in synaptic plasticity and tau phosphorylation (37).

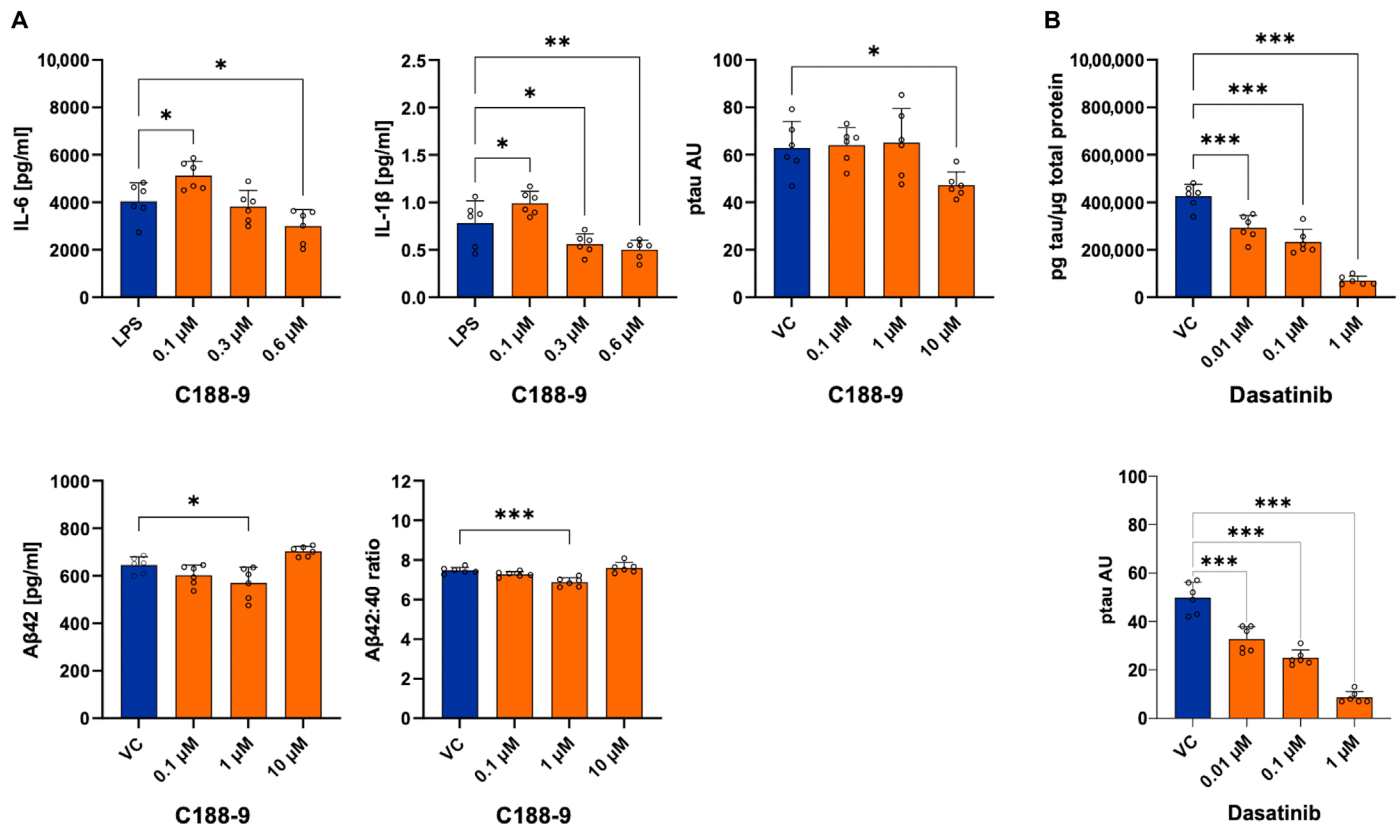


Fig. 9. Phenotypic screening of candidate AD drug targets. (A) STAT3 inhibitor C188-9 rescued three AD phenotypes: lipopolysaccharide (LPS)–induced neuroinflammation [interleukin-6 (IL-6) and IL-1β secretion], tau phosphorylation (ptau), and Aβ secretion (Aβ42 and Aβ42:Aβ40). C188-9 reduced IL-6 relative to LPS (blue bar) at the highest concentration (0.6 μM) and reduced release of IL-1β relative to LPS at both 0.3 and 0.6 μM concentrations. For both cytokines, there was a significant increase relative to LPS at the lowest concentration (0.1 μM). C188-9 significantly reduced levels of phosphorylated tau (ptau) relative to VC (blue bar) at the highest concentration (10 μM). C188-9 significantly reduced endogenous Aβ1–42 and Aβ42:Aβ40 ratio relative to VC (blue bar) at the middle concentration (1 μM). (B) The YES1/FYN inhibitor dasatinib reduced levels of total tau and ptau relative to the VC (blue bar) at all three concentrations (0.01, 0.1, and 1 μM). Pro-inflammatory cytokines IL-6 and IL-1β (pg/ml) were measured in the supernatant of BV2 (microglial) cells after 24-hour LPS stimulation and assessed using MSD V-PLEX. Levels of tau (pg tau/μg of total protein) and ptau [arbitrary units (AU)] were measured in lysates from SH-SY5Y cell line overexpressing mutant human tau441 (SH-SY5Y-TMHT441) after 24 hours of stimulation. Levels of Aβ42 and Aβ40 (pg/ml) were measured in the supernatant of murine BV2 (microglial) cells after 3 hours of Aβ stimulation in human APP-overexpressing H4 neuroglioma cells. Blue bars indicate the comparison group: either LPS (stimulation to generate proinflammatory cytokines) for LPS-induced neuroinflammation or the VC for tau phosphorylation and Aβ secretion. Orange bars indicate three increasing concentrations for treatment with C188-9 or dasatinib. Values were compared by one-way ANOVA followed by Dunnett's multiple comparison test, and significant differences were indicated (* $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$). LPS, lipopolysaccharide; VC, vehicular control (0.1% DMSO); ptau, phosphorylated tau231.

The incipient AD signature also contains several proteins with previously unknown roles in AD pathogenesis that may mediate biological actions relevant to AD. These include methionine aminopeptidase 1 (METAP1), which catalyzes removal of N-terminal methionine from newly synthesized proteins and plays an important role in cell cycle progression (38); the chemokine CCL19; and interferon-λ (IFN-λ), a member of the interferon family that has been shown to inhibit infection of primary neurons and astrocytes by neurotropic viruses (39). Together with altered levels of STAT3, a key signal transducer of cytokine signaling, these findings suggest an early involvement of neuroinflammation in AD progression. The presence of galectin-8 (LGALS8), a β-galactoside-binding lectin in the incipient AD signature, suggests converging pathways between host defense mechanisms against microbial infection and neurodegeneration (40). Recent evidence also suggests that LGALS8-mediated autophagy is important in preventing the entry

of tau seeds into the cytosol and their subsequent aggregation (41). Our immunohistochemical studies demonstrate that LGALS8 is localized within neuronal lysosomes where it may play a role in autophagic degradation of intraneuronal tau (42). Our finding that dual-specificity phosphatase 3 (DUSP3), also known as VH1-related phosphatase (VHR), is localized within postsynaptic processes of both excitatory and inhibitory neurons is also consistent with its proposed role in countering excitotoxicity-induced neuronal death and Aβ accumulation (43). Table S5 summarizes key biological roles of proteins in the incipient AD signature.

As seen in Fig. 2D, intriguingly, for nearly all proteins in the incipient AD proteomic signature, the direction of association was opposite between young *APOE ε4* carriers and older AD individuals, with most proteins being elevated in young *APOE ε4* carriers relative to noncarriers and reduced in AD relative to CN samples. This suggests that alterations in brain proteomic profiles in young *APOE*

$\epsilon 4$ carriers may represent early pathogenic changes in individuals at enhanced risk for AD. Over the next three to four decades, the functional consequences of these early molecular changes may cause the progressive accumulation of pathology, eventually manifesting as the irreversible cognitive impairment and functional decline characterizing the clinical syndrome of AD. The associations of brain tissue levels of several of the proteins identified with both severity of AD pathology and ante-mortem trajectories of cognitive performance in AD individuals in both BLSA and ROS add further strength to the hypothesis that early perturbations in biological pathways represented by these proteins may be causally associated with AD progression. This hypothesis merits further testing in experimental AD models.

The opposite direction of associations between young *APOE* $\epsilon 4$ carriers and AD individuals lends itself to multiple interpretations that merit further exploration. This may be consistent with a possible antagonistic pleiotropy effect of *APOE* $\epsilon 4$, whereby some of the biological changes associated with the $\epsilon 4$ allele confer selective advantages early in life but turn detrimental with age (44). Others have suggested that an evolutionary or environmental mismatch may explain the effects of the $\epsilon 4$ allele, as multiple lines of evidence have indicated that *APOE* $\epsilon 4$ promotes a pro-inflammatory, highly responsive innate immune response (45) that has been shown to protect multiple health indices, including cognition, in regions with high prevalence of infectious disease (46). Studies among the Bolivian Tsimane have demonstrated that the *APOE* $\epsilon 4$ allele protects against cognitive decline only among those with a high parasite burden but contributes to more rapid decline in those without high parasite loads (47). Such a mismatch hypothesis appears plausible in the context of our study, in which multiple proteins in the incipient AD signature have defined roles in the innate immune system and related roles in signal transduction (table S5).

Although differential expression of individual proteins in brain tissue samples may provide novel insights into AD pathogenesis, we were also interested in the plausible functional consequences of alterations in the biological pathways represented by the proteins we measured. Our GSEA results therefore provide additional biological context to our findings by revealing alterations in numerous molecular pathways in young *APOE* $\epsilon 4$ carriers relative to non-carriers. Several of these pathways are also altered in AD brain relative to CN samples. Similar to our findings of alterations in individual protein levels described above, our GSEA analyses reveal several biological functions/pathways that appear to be enriched in young *APOE* $\epsilon 4$ carriers relative to noncarriers (YAPS) and diminished in AD relative to CN individuals (BLSA/ROS). These include signaling cascades involving epidermal growth factor (EGF), insulin, G protein-coupled receptors (GPCRs), diacylglycerol/inositol trisphosphate/calcium (DAG/IP3/Ca²⁺), neurotransmitters, opioids, and peptide hormones. Other enriched modules include endothelial cell function, axon guidance, protein autophosphorylation, and humoral immune responses.

Secondary validation of our index results from aptamer-based proteomics within multiple independent samples, and orthogonal assays are a key strength of our study. By comparing aptamer-based brain proteomic findings in BLSA, ROS, and YAPS with deep MS-based proteomic profiling results in the Mount Sinai cohort, we were able to validate our results using an independent sample of AD and CN participants. Then, using the 5xFAD model, we were able to assess alterations in brain protein levels in young *APOE* $\epsilon 4$ and AD individuals that may be driven by aberrant APP processing (i.e.,

convergently altered in 5xFAD transgenic mice) and those likely to be independent of A β accumulation (i.e., not altered in 5xFAD transgenic mice). Furthermore, assessment of differentially expressed gene transcripts encoding these proteins within excitatory and inhibitory neurons provides additional functional context to our results. Several of the incipient AD signature proteins were also reported in Johnson *et al.*'s (48) recent brain proteomic study of AD, of which DUSP3, HMOX2, KPNB1, and STAT3 were also found to be differentially abundant in AD compared to control individuals (see table S16).

By accessing the YAPS cohort, we were able to focus on *APOE*-related proteomic alterations in young individuals decades before the typical age at onset of AD, thereby allowing us to relate these changes to AD pathogenesis at the early preclinical stages of disease progression. The main limitation of our study is the cross-sectional nature of the majority of our analyses, which precludes testing temporal relationships between brain proteomic alterations and AD progression. Furthermore, while we used the SomaLogic 1.3K platform to enable independent confirmation of our results by other groups (49), our results are limited to only proteins included on this platform. Future studies using unbiased proteomics with a larger coverage of the proteome may uncover additional alterations in other molecular pathways in AD.

Some consideration of the demographic characteristics of the BLSA and ROS cohorts is also important in the interpretation of our results. Key demographic differences between BLSA and ROS include a predominantly female sample in ROS, whereas less than half of BLSA participants included in this report were female. Previous neuropathological studies in ROS have established that women have a greater global burden of AD pathology and neurofibrillary tangle density in the brain as well as more severe arteriosclerosis (50). AD participants in the BLSA also had an earlier onset and longer duration of disease compared to those in ROS. These differences may partly explain regional variations in the distribution of protein differences between AD and CN samples in the ITG/MFG across the two cohorts.

Another important consideration is the convergence of our findings across diverse samples, including multiple human autopsy cohorts, two transgenic AD mouse models, and single-cell transcriptomic datasets. While we observe that the majority of proteins in the incipient AD signature are also differentially abundant between AD/CN and or Tg/WT protein/transcript samples in at least two independent validation datasets, there are inconsistencies in the direction of observed differences. This likely reflects significant heterogeneity in these models, including differences in disease severity and stage of disease progression, as well as limitations of transgenic models in recapitulating all aspects of AD pathogenesis and the complex gene-environment interactions driving AD progression in humans. Similarly, we have applied three methodologically distinct approaches to detect and quantify protein levels in this study, i.e., aptamer- and MS-based proteomics as well as antibody-based Western blotting. Fundamental differences in these methods may also contribute to inconsistencies in directionality of some of the observed changes in protein levels. For example, antibody- and aptamer-based methods rely upon access to specific epitopes on a given protein for its detection and posttranslational modifications at these epitopes may affect the signal arising from these assays. Previous studies comparing consistency of results across these various proteomic platforms have shown that there is only a modest convergence of results from these distinct approaches (51, 52).

In summary, the goal of this study was to examine molecular correlates of *APOE*-related AD risk before accumulation of AD pathology and preceding the onset of clinical symptoms. We have performed a proteomic study of human brain tissue samples in multiple human cohorts and in two transgenic AD mouse models to identify early protein markers of AD and dysregulation in several biological pathways in *APOE* $\epsilon 4$ carriers that may be causally related to the eventual accumulation of neuropathology and symptom onset in AD. Some of the proteins implicated include those targeted by existing and experimental drugs in other diseases. Phenotypic screening of drugs targeting proteins in the incipient AD signature suggests that these molecular pathways may represent novel therapeutic targets. These findings may pave the way for future studies to fully understand the biological basis of *APOE*-associated AD risk and to develop effective interventions targeting the earliest molecular drivers of the disease.

MATERIALS AND METHODS

Stage 1: Discovery of the incipient AD signature

Subject details

Baltimore Longitudinal Study of Aging. The National Institute on Aging's (NIA) BLSA is among the longest running scientific studies of aging in the United States (53). This observational study began in 1958 and includes longitudinal, radiological, clinical, and laboratory evaluations of community-dwelling volunteer participants. The individuals included in this study were participants in the autopsy substudy of the BLSA, which has been described previously (54). Postmortem brains were examined by an expert neuropathologist to assess AD pathology. The CERAD and Braak criteria were used to assess severity of AD pathology based on neuritic plaques (14) and neurofibrillary tangles (55), respectively, as described previously (56). Clinical diagnoses of dementia and AD have previously been described (57) and were based on the Diagnostic and Statistical Manual (DSM)–III-R (58) and National Institute of Neurological and Communication Disorders and Stroke–Alzheimer's Disease and Related Disorders Association (NINCDS-ADRDA) criteria, respectively (59).

Autopsy participants were classified as either AD or CN according to the following criteria: AD participants ($n = 31$) had a clinical diagnosis of AD or mild cognitive impairment (MCI) within 1 year of death in addition to a postmortem CERAD pathology score > 1 (i.e., CERAD B or C), and CN participants ($n = 19$) had normal cognition within 1 year of death and a CERAD pathology score ≤ 1 (i.e., CERAD 0 or A). Diagnosis and cognitive status were determined at consensus diagnosis conferences using procedures described in detail previously (57). Demographic characteristics of the BLSA cohort are included in Table 1. The BLSA study protocol has ongoing approval from the Institutional Review Board of the National Institute of Environmental Health Science, National Institutes of Health (NIH).

Religious Orders Study. The ROS has enrolled Catholic nuns, priests, and brothers from a multitude of communities across the United States since 1994 (60). This longitudinal observational study collects information from clinical, neuroimaging, laboratory, and self-report evaluations of employed and retired community-dwelling individuals. At the time of enrollment, participants did not have a diagnosis of known dementia. All participants agreed to organ donation and annual clinical evaluation.

Our sample consisted of a subset of participants from the larger ROS cohort study. All ROS participants provided written informed consent, and the study was approved by an Institutional Review Board of the Rush University Medical Center. Participants signed an Anatomical Gift Act for organ donation and a repository consent to allow their data and biospecimens to be shared. At each study visit, dementia status was determined by trained clinicians using all cognitive and clinical data blinded to previous years based on NINCDS-ADRDA criteria. A final consensus clinical diagnosis was determined at death blinded to all neuropathologic data. Autopsies were performed on the basis of standard methods reported previously (61). Postmortem brains were examined by an expert neuropathologist or trained technician to assess AD pathology. CERAD and Braak criteria were used to assess severity of AD pathology, as described previously (62).

Participants were classified into two groups. AD participants ($n = 31$) had a final clinical diagnosis of AD and a NIA-Reagan score of intermediate or high likelihood of AD. NIA-Reagan criteria are based on both neuritic plaques (CERAD score) and neurofibrillary tangles (Braak score) (63). CN participants ($n = 22$) had a clinical diagnosis of no cognitive impairment and a NIA-Reagan score of low likelihood of AD or no AD. Diagnosis and cognitive status were determined on the basis of a three-stage process described previously (60). Demographics of the ROS autopsy cohort are included in Table 1.

Young *APOE* Postmortem Study. The YAPS was composed of postmortem brain tissue samples acquired from the Brain Resource Center at the Johns Hopkins Alzheimer's Disease Research Center (ADRC). All autopsies were performed at the OCME of the State of Maryland in Baltimore. Study of postmortem samples was conducted under a protocol authorized by the Johns Hopkins University Institutional Review Board. The clinical and cognitive status of these subjects was undetermined. Brain tissue samples from these participants had CERAD and Braak scores of 0, indicating the absence of substantial AD pathology. See Pletnikova *et al.* (7) for additional details on the study sample. YAPS samples included in this study are a convenience sample designed to include approximately 50% *APOE* $\epsilon 4+$ individuals ($n = 18$ $\epsilon 4+$ and $n = 17$ $\epsilon 4-$). *APOE* genotyping was conducted on frozen tissue using the methods of Hixson and Vernier (64). Demographic characteristics of the YAPS cohort are included in Table 1. The age of individuals included in the YAPS cohort precedes the typical age of onset of AD by approximately three decades (65).

Aptamer-based proteomics

Brain tissue collection and homogenization. YAPS, BLSA, and ROS brain tissue samples were selected from two a priori specified regions: the ITG and MFG, which are susceptible to accumulation of classical AD neuropathology. All brain samples were stored at -80°C . For sample extraction, brain samples were placed in -20°C freezer for 15 min and then sterile, 4-mm-diameter tissue punches were used to extract samples from the cortical surface of the brain tissue regions. Samples were again stored at -80°C before proteomic assays.

To ~ 10 mg of brain tissue, 110 μl of T-PER (tissue protein extraction reagent) (Thermo Fisher Scientific, USA) with 2 μl of Halt Protease and Phosphatase inhibitor cocktail (Thermo Fisher Scientific, USA) was added and placed in a CKMix grinding tube, containing soft tissue homogenizing lysis beads [a mix of 1.4-mm and 2.8-mm ceramic (zirconium oxide) beads] (Bertin Technologies, San Quentin, France). The tubes were placed in a Precellys Evolution

tissue homogenizer (Bertin Technologies, San Quentin, France) and homogenized for two 30-s cycles of 6500 rpm and a 30-s rest in between. The homogenate was removed and placed in an Eppendorf tube and centrifuged at 16,000g for 5 min at 4°C. The supernatant was removed and centrifuged a second time for 10 min at 16,000g at 4°C. The supernatant was collected at 4°C, 2.5 μ l was aliquoted, and protein quantitation was carried out using the MicroBCA Protein Assay Kit (Thermo Fisher Scientific, USA). The total protein concentration was determined, and the samples were diluted to a final volume of 200 μ g/ml with 1 $\times$ phosphate-buffered saline (PBS) and stored at –80°C until analysis.

Proteomic quantification. Sample total protein was adjusted to 16 μ g/ml in SB17T buffer (40 mM Hepes, 125 mM NaCl, 5 mM KCl, 5 mM MgCl₂, 1 mM EDTA, and 0.05% Tween 20 at pH 7.5). Proteomic profiles for 1322 SOMAmers were assessed using the 1.3K SOMAscan assay at the Trans-NIH Center for Human Immunology and Autoimmunity, and Inflammation (CHI), National Institute of Allergy and Infectious Disease, NIH (Bethesda, MD, USA). The SOMAscan assay platform includes 1322 SOMAmer Reagents, of which 12 are hybridization controls, 5 are viral proteins, and 5 are nonspecifically targeted SOMAmers. As a result, analyses included 1300 SOMAmer Reagents. The proteins for which SOMAmers included in this assay were selected are included in table S1. The experimental procedure for proteomic assessment and normalization has been previously reported (66). In brief, targets were generated by a process known as Selected Evolution of Ligands by Exponential Enrichment (SELEX), a method of identifying high-affinity binding targets from much larger sequence libraries. This method allows the accurate detection of proteins spanning a dynamic range of eight orders of magnitude (67). The SOMAscan platform has been used widely for proteomic quantification in the context of multiple diseases and tissues, allowing replication and validation of findings.

The SOMAscan assay uses SOMAmers to translate protein concentrations into measurable DNA signals, which can be quantified using standard DNA detection procedures. This is achieved by affinity binding and biotin capture on streptavidin beads. The DNA concentrations obtained from this method are reported as relative fluorescence units (RFUs), resulting from fluorescent SOMAmer hybridized to its complimentary probe on an Agilent array, and are directly proportional to the reported relative abundance of SOMAmer Reagents. Study/cohort-specific samples were run in the same batch on separate plates. Within study/cohort, samples were randomized by disease (AD and CN), brain region (ITG and MFG), sex, and age. The data normalization process across all plates and cohorts includes hybridization, control normalization, median signal normalization, and calibration normalization, as previously described (66). As an independent quality check (QC) for each analyte, we determined the coefficient of variation (median, 13 to 15%) obtained from four pairs of technical duplicates according to three different methods (68–70). For more details on the assay's performance, see Candia *et al.* (66). Please see Supplementary Text and table S17 for additional details on measurement range and reproducibility/replicates.

Statistical analyses

Demographic characteristics. Demographic characteristics of the three cohorts (YAPS, BLSA, and ROS) are summarized in Table 1. Comparisons between BLSA and ROS cohorts were performed using two-sample *t* tests for continuous variables and χ^2 tests for independence for categorical variables.

Incipient AD signature: Definition and analysis. We undertook a multistep process to identify proteins significantly altered in all three primary cohorts in this study (i.e., BLSA, ROS, and YAPS). In step 1, we defined the AD proteomic signature as the proteins differentially abundant between AD and CN in both BLSA ($n_{AD} = 31$ and $n_{CN} = 19$) and ROS ($n_{AD} = 31$ and $n_{CN} = 22$), i.e., shared proteins in two independent cohorts, in either the ITG or MFG. We used separate proportional odds models (71), a generalization of the nonparametric Wilcoxon and Kruskal-Wallis tests, for each protein in both the ITG and MFG brain regions. All models included ranked protein levels (outcome), the group predictor (AD versus CN), and covariates—sex and age at death. Statistical significance of differentially abundant proteins included in the AD proteomic signature was corrected for multiple comparisons using a Benjamini-Hochberg FDR (72)—adjusted $P < 0.10$ in both BLSA and ROS.

In step 2, we then determined whether proteins included in the AD proteomic signature were also significantly different between young *APOE* $\epsilon 4$ carriers ($n = 18$) and noncarriers ($n = 17$) in either the ITG or MFG in the YAPS cohort. Participants were indicated as *APOE* $\epsilon 4+$ if they had at least one $\epsilon 4$ allele; participants were indicated as *APOE* $\epsilon 4-$ if they did not carry the $\epsilon 4$ allele. Separate proportional odds models were performed as above, including the ranked protein levels (outcome), the group predictor ($\epsilon 4+$ versus $\epsilon 4-$), and covariates—sex, race (white versus non-white), and age at death. Because analyses were restricted only to those proteins identified as dysregulated in the AD proteomic signature, the significance threshold was set as $P < 0.05$ for identifying proteins differing in the YAPS cohort. The incipient AD signature was defined as the proteins in the AD proteomic signature that were also altered in $\epsilon 4$ carriers versus noncarriers in the YAPS cohort.

Controlling for type I error. In the “Stage 1: Discovery” section, we derived a set of proteins of interest by identifying proteins that differed significantly between conditions in each of the three discovery cohorts. We addressed the issue of potential type I errors in two ways. First, in the first step of the discovery stage where the AD proteomic signature was defined, we corrected for multiple comparisons using an FDR threshold. In the second step of the discovery stage where we defined the overlap between the AD proteomic signature and proteins differentially expressed by *APOE* carriers in YAPS (i.e., the incipient AD signature), we computed the statistical distributions of the multiset intersection and calculated their exact probabilities to determine the statistical significance of the intersection between YAPS, BLSA, and ROS using the SuperExactTest package in R (11). This method determines the number of intersecting proteins that would be expected across the three cohort signatures if the proteins were picked at random and compares the null “expected intersection” to the “observed intersection” to generate a *P* value and fold enrichment (i.e., observed intersection size/expected intersection size).

Second, we applied a rigorous two-step validation approach in the “Stage 2: Validation” section to confirm our index results (i.e., the incipient AD proteomic signature). First, in the “Step 6a: Primary validation” section, we selected a subset of six proteins from the incipient AD signature for validation based on their plausibility as targets of approved and experimental drugs in other diseases. We validated these proteins using immunoblotting of brain homogenates from the 3xTg-AD transgenic mouse model of AD. Next, in the “Step 6b: Secondary validation” section, we validated the incipient AD signature in three independent publicly available datasets

using orthogonal methods, including (i) two-dimensional liquid chromatography–tandem MS (LC/LC-MS/MS)–based proteomics in the Mount Sinai Brain Bank and a 5xFAD transgenic mouse model of AD, and (ii) single-cell transcriptomics (i.e., scRNA-seq) from the ROSMAP cohort. These validation studies are described in detail below in the “Stage 2: Validation of the incipient AD signature” section.

Pathway enrichment and protein-protein interaction analysis. To determine whether proteins in the incipient AD proteomic signature were functionally related, we performed pathway enrichment analysis using the MSigDB database (<https://gsea-msigdb.org/gsea/msigdb/>) (73) and protein-protein interaction analyses using StringDB (<https://string-db.org/>) (74).

StringDB uses numerous publicly available sources to provide both physical and functional interactions between proteins, further visualizing results as a nodal interaction network (74). We input the 25 proteins from the incipient signature to the StringDB database and recorded the internode interactions resulting from the program’s computational predictions.

Associations with AD pathology. In step 3, we tested whether brain tissue protein levels in the incipient AD proteomic signature were associated with severity of AD pathology in BLSA and ROS within the ITG and MFG. Similar to previous studies (75), we examined partial Spearman correlations of CERAD and Braak scores with ranked aptamer values, controlling for covariates—mean-centered sex and age at death. A significant ($P < 0.05$) positive correlation indicated that higher concentration of the protein was associated with higher AD pathology (i.e., higher CERAD or Braak scores), and a significant ($P < 0.05$) negative correlation indicated that lower concentration of the protein was associated with higher AD pathology.

Associations with longitudinal cognitive performance. In step 4, we tested whether brain tissue protein levels in the incipient AD proteomic signature were associated with antemortem trajectories of cognitive performance among individuals with AD in BLSA and ROS. Similar to previous studies (71), linear mixed-effects models were used to determine whether brain tissue protein levels at death were associated with longitudinal changes in cognitive performance, specifically the MMSE, before death. MMSE scores at each visit were used as the outcome variable. Predictors included protein, sex, age at death, time (time to the last visit), protein * time, age at death * time, and sex * time. Random effects included a random intercept. The origin of time variable was anchored to the last visit. The coefficient of interest was protein * time: A significant ($P < 0.05$) positive coefficient indicated that higher concentration of the protein was associated with slower/reduced decline in MMSE over time; a significant ($P < 0.05$) negative coefficient indicated that increased concentration of the protein was associated with faster increased decline in MMSE over time. Similar to previous analyses (76), protein concentration (predictor) was centered at the median, rescaled using the interquartile range (IQR), and outliers greater than 3 * IQR were excluded. Age of death was mean-centered, and sex was coded 0 as male and 1 as female.

Gene set enrichment analysis. In step 5, we used GSEA to identify the AD-related biologic pathways that may be altered in young *APOE* $\epsilon 4$ individuals by comparing enriched gene sets in the young sample to the older adult sample. GSEA offers an important complement to analyses of differences in abundance of individual proteins by determining the extent to which biologically defined

collections of genes are affected as a group under a given biological context (73). GSEA captures groups of proteins that share common biologic functions that may be different between AD and CN despite nonsignificant differences in single-protein analyses. We performed GSEA in R using the *fgsea* package (77) on all 1300 proteins included in our dataset and selected gene sets from the Molecular Signatures Database (v6.0 MsigDB). We conducted exploratory GSEA analysis using the following gene sets from MsigDB: 4 from the Blalock *et al.* AD gene sets (78), 289 from BioCarta, 186 from Kyoto Encyclopedia of Genes and Genomes (KEGG), 1499 from Reactome, and 7350 from Gene Ontology (GO) Biological Processes. We excluded gene sets with <10 and >300 genes, resulting in 2406 gene sets used in GSEA. In these analyses, significance was set as an FDR-corrected $P < 0.05$. Proteins were ranked for GSEA based on the odds ratio (OR) calculated by the proportional odds models described in step 1. In GSEA analyses, magnitude of enrichment of gene sets was quantified using the NES. The NES represents a weighted Kolmogorov-Smirnov test statistic and corresponds to the extent to which a specific gene set is overrepresented at the top or bottom extremes of the ranked protein list. A positive NES represents overexpression of a gene set, while a negative NES represents underexpression.

To determine the statistical significance of the overlap between AD and YAPS enriched gene sets, we tested the significance of this overlap by performing Fisher’s exact ratio tests between the number of overlapping gene sets at FDR thresholds of 0.05, 0.1, and 0.25.

Sensitivity analyses. To test whether proteomic differences between AD and CN in the older cohorts were driven predominantly by *APOE* $\epsilon 4$ individuals, we excluded all *APOE* $\epsilon 4+$ individuals from ROS and BLSA and performed proportional odds models identical to the main analysis. We then examined whether proteins included in this non-*APOE* $\epsilon 4+$ AD proteomic signature differed between *APOE* $\epsilon 4$ carriers and noncarriers in the YAPS cohort as in the main analysis.

Similarly, we performed additional sensitivity analyses in which we restricted the sample to only *APOE* $\epsilon 4$ carriers in ROS and BLSA and compared AD and CN individuals. Owing to small sample size, we performed proportional odds models similar to the main analysis without covariates and only tested proteins included in the incipient AD proteomic signature.

Stage 2: Validation of the incipient AD signature

To validate the incipient AD proteomic signature identified in the “Stage 1: Discovery” section, we undertook a two-step validation of results. In the “Step 6a: Primary validation” section, we performed Western blotting of brain homogenates from 3xTg-AD mice and human samples (BLSA MFG) and assessed the subcellular localization of these proteins in the human brain using immunohistochemistry. In the “Step 6b: Secondary validation” section, we validated the signature using orthogonal methods, i.e., MS-based proteomics in (i) AD and CN samples in the Mount Sinai Brain Bank, (ii) 5xFAD transgenic mouse model of AD, and (iii) scRNA-seq from brain samples in the ROSMAP.

Step 6a: Primary validation

Western blot in 3xTg-AD mouse model and human brain samples. We selected a subset of six proteins from the incipient AD proteomic signature for Western blot validation. We first assayed these proteins in a 3xTg-AD mouse model. This model expresses mutant forms of human APP, presenilin-1, and tau, developing age-dependent accumulation of extracellular A β plaques, intracellular tau accumulation, oxidative stress, and cognitive deficits (12). Proteins were

chosen for validation because of previous evidence suggesting that they are targeted by approved and experimental drugs for other diseases (table S9) and may therefore present plausible novel drug repurposing opportunities in AD.

Age-matched (young and old) transgenic ($n = 6$) and WT ($n = 6$) male and female adult mice were used in all experiments. Mice were maintained on a standard NIH diet ad libitum in a 12-hour light/dark cycle. All mice were housed in the NIA, Baltimore. The animals were group-housed where possible. All animal experiments were performed using protocols approved by the appropriate institutional animal care and use committee of the NIA.

We performed identical Western blot analyses in the MFG from human brain samples in the BLSA AD ($n = 11$) and CN ($n = 11$) individuals. These samples were a convenience subset of available BLSA samples used in the “Stage 1: Discovery” section. Demographic characteristics of the sample are described table S10.

For both mouse and human tissues, brain homogenates were solubilized in 1× radioimmunoprecipitation assay (RIPA) buffer containing protease inhibitor cocktail (Roche). Standard Western blot procedures were followed. Equal amount of proteins was resolved on 4 to 20% gradient gels (Bio-Rad) using SDS–polyacrylamide gel electrophoresis (SDS–PAGE). Proteins were transferred to 0.2- μ m pore size polyvinylidene difluoride (PVDF) membrane using the wet transfer system (Bio-Rad). Membranes were blocked with 3% milk (Bio-Rad) at room temperature (RT) for 1 hour and then incubated with primary antibodies overnight at 4°C. The following primary antibodies were used: STAT3 (Cell Signaling Technology, 30835), LGALS8 (Novus, NBP2-75501), TOP1 (Thermo Fisher Scientific, MA5-32228), DUSP3 (ABclonal, A12068), FYN1 (Invitrogen, MA1-19331), and YES1 (Proteintech, 20243-1-AP). Secondary horseradish peroxidase–conjugated antibodies and ECL prime (GE Healthcare Bio-Sciences) or SuperSignal West Femto Chemiluminescent Substrates (Thermo Fisher Scientific) were used to visualize signals on a ChemiDoc XRS+ system (Bio-Rad Laboratories, Hercules, CA, USA). β -Actin was used for the loading control and normalization for total brain lysates. Digitized images were obtained, processed, and quantified with ImageLab version 6.1 (Bio-Rad Laboratories).

Immunohistochemical staining of human brain tissue samples. The autopsied brain samples were obtained at the OCME of the State of Maryland in Baltimore, and brains were accessioned as previously described (13). We followed protocols authorized by the Institutional Review Board (IRB) of the State of Maryland Department of Health and Human Services and Johns Hopkins Medicine. For the primary screen to determine whether proteins showed positive immunoreactivity, we used at least two commercially available antibodies for each selected protein (see table S12). Proteins that screened immunoreactivity positive were then selected for secondary screening for subcellular localization.

All tissue sections were deparaffinized in xylene and rehydrated in 100 and 95% EtOH. To ascertain good preservation of tissues to be examined, we screened sections of the cerebellum with β -tubulin immunohistochemistry as previously described (13). Antigen retrieval was performed using HistoVT (Nacalai) or Dewax and HIER Buffer M (Eppredia) at 95°C for 20 min.

After washing with tap water, antigen retrieval was performed in 1 mM EDTA (pH 8.0) (Invitrogen, 15575-038) by boiling for 4 min. Then, all samples were blocked in PBS with 5% normal goat serum (Sigma-Aldrich), 5% normal donkey serum, and 0.2% Triton X-100 for 1 hour at RT. Primary antibodies were incubated in blocking

buffer for 16 hours at 4°C. On the following day, samples were washed in PBS for 5 min $\times$ 3, and then we applied Alexa Fluor secondary antibodies in PBS with 0.5% Tween 20 and incubated for 1 hour at RT. Samples were washed in PBS once for 5 min, and then Hoechst 33258 (5 μ g/ml) in PBS was applied and incubated for 20 min at RT. Samples were washed in PBS once for 5 min. For quenching lipofuscin autofluorescence, we used TrueBlack Lipofuscin Autofluorescence Quencher (Biotium, 23007) diluted 1:40 in 70% EtOH, applied to the samples, and incubated for 50 s at RT. To facilitate the TrueBlack reaction, samples were constantly swirled by hand during the incubation. Then, samples were washed in PBS for 5 min $\times$ 3 and coverslipped using ProLong Gold Antifade reagent (Invitrogen, P36930). To enhance SNR of synaptic protein immunoreactivity, we used an ultrafast optical clearing method solution (79) mixed with a commercial antifade mounting solution. Stained tissue sections were kept at 4°C at least 2 days before imaging. Immunofluorescent images were taken on a Zeiss LSM 700 confocal microscope in the Microscope Facility of the Johns Hopkins School of Medicine.

Step 6b: Secondary validation subject details and data acquisition

All secondary validation analyses on the publicly available datasets used were performed in collaboration with corresponding authors from the index publications and described below. Relevant analytic code and guidance on data use were requested directly from study authors.

Mount Sinai Brain Bank. We obtained proteomic data from Brodmann area 36 brain region samples from the Mount Sinai Brain Bank sample comparing AD ($n = 39$) and CN ($n = 23$) individuals. These data were recently published and are available from the AMP-AD Knowledge Portal (<https://adknowledgeportal.synapse.org>) (15). Proteins were quantified using acidic pH reversed-phase LC-MS/MS. Demographic information for this cohort is available in the Supplementary Materials from the source publication. AD diagnosis in this cohort was made using cognitive and neuropathological assessments as described previously (80). Previous studies have indicated an appropriate level of complementarity between LC-MS/MS and SOMAscan methods such that high degrees of concordance in proteomic quantification may be obtained across the two platforms. Differences in results across platforms have primarily been attributed to posttranslational modification enrichment, suggesting that such modifications may better explain divergence of results across the two platforms than differences in protein abundance (52).

5xFAD mouse model. Proteomic data were obtained from cortical brain samples from the 5xFAD transgenic mouse model of AD comparing transgenic ($n = 4$) to WT ($n = 4$) mice (15). These data were published recently and are accessible at the Proteomics Identification Database (PRIDE; <https://ebi.ac.uk/pride/>). Protein quantification was achieved by acidic pH reversed-phase LC-MS/MS. In this study, analyses were conducted at the 6-month time point, which is associated with substantial AD pathology and memory impairment in the 5xFAD mouse model (17).

ROSMAP scRNA-seq. scRNA-seq gene expression data (16) from the ROSMAP (60) were downloaded from Synapse (<https://synapse.org/#!Synapse:syn18485175>). Code used to run the analyses of Mathys *et al.* (16) was requested from coauthors. Postmortem data were collected from ROSMAP participants: 32 individuals (18 male and 14 female) in the AD category and 14 individuals (5 male and 9 female) in the CN category. The AD category included individuals with a clinical diagnosis of AD, as well as individuals with a clinical

diagnosis of MCI and no other condition contributing to cognitive impairment. The CN category included individuals with a clinical diagnosis of no cognitive impairment.

Tissue was profiled from the prefrontal cortex (Brodmann area 10) across eight major cell types in the aged dorsolateral prefrontal cortex: inhibitory neurons, excitatory neurons, astrocytes, oligodendrocytes, microglia, oligodendrocyte progenitor cells, endothelial cells, and pericytes. Additional details are provided in the index paper (16). In our validation studies, we restricted analyses to inhibitory and excitatory neuron cell types in which a majority of transcripts for proteins in the incipient AD signature were quantified.

Statistical analyses

3xTg-AD mouse model. In the 3xTg-AD transgenic mouse model of AD, we tested whether proteins were differentially expressed between young and old transgenic and WT mice. Protein expression was calculated as the ratio of protein staining intensity to its corresponding β -actin intensity. We used two-sample *t* tests (parametric) to calculate differences in protein expression between 3xTg-AD transgenic and WT mice. We additionally used the Wilcoxon rank sum test (non-parametric) to confirm that results were robust to distributional assumptions. Significant differences were indicated as $P < 0.05$.

Mount Sinai Brain Bank. In the Mount Sinai Brain Bank sample, we tested whether proteins were differentially abundant between AD and CN samples. Twenty one of 25 proteins from the incipient AD signature were quantified by MS-based proteomics. Protein data were previously corrected for age and sex. We performed a one-way test for differential abundance between the AD and CN samples using the R package limma (81). Significance was indicated as $P < 0.05$.

5xFAD mouse model. In the 5xFAD transgenic mouse model of AD, we tested whether proteins were differentially abundant between 5xFAD transgenic and WT mice. Twenty one of 25 proteins from the incipient AD signature were quantified in the cortex. We performed two-tailed Student's *t* tests. Protein levels were analyzed at the 6-month time point, and significance was indicated as $P < 0.05$.

ROSMAP scRNA-seq. In the ROSMAP scRNA-seq sample, we tested whether mRNA levels of the genes associated with the proteins in the incipient AD signature were differentially abundant between AD and CN samples. Twenty two of 25 gene transcripts in both inhibitory and excitatory neurons from the proteins in the incipient AD signature were quantified. We scaled each sample to have the same total read count (82). To test differences between AD and CN, we performed Wilcoxon rank sum tests. Similar to the source publication (16), each single cell-specific sample from a participant was treated as an independent sample. We compared transcript levels for excitatory and inhibitory cell types between AD and CN separately, and significance was indicated as $P < 0.05$.

Stage 3: Phenotypic screening of approved drugs targeting selected proteins in the incipient AD signature

We tested whether approved/experimental drugs targeting proteins in the incipient AD signature could rescue distinct molecular phenotypes relevant to AD without adverse effects on cell viability. We tested three drugs known to target STAT3 (crizotinib, napabucasin, and C188-9) and one drug targeting YES1 and FYN (dasatinib). We selected STAT3 and the Src family tyrosine kinases YES1 and FYN in these experiments as they have been extensively studied as drug targets in cancer. Our choice of candidate AD treatments in these experiments was based on the availability of FDA-approved drugs used in current clinical practice that target STAT3 (e.g., crizotinib)

or YES1 and FYN (e.g., dasatinib) or experimental drugs currently being tested in clinical trials (e.g., napabucasin and C188-9). Drug concentrations tested are included in table S15.

LPS-induced neuroinflammation

The murine microglial cell line BV2 was cultivated in Dulbecco's modified Eagle's medium (DMEM) medium supplemented with 10% fetal calf serum (FCS), 1% penicillin/streptomycin, and 2 mM L-glutamine (culture medium). For LPS stimulation assay, 5000 BV2 cells per well (uncoated 96-well plates) were plated out and the medium was changed to treatment medium (DMEM, 5% FCS, and 2 mM L-glutamine). After changing cells to treatment medium, drug compounds were administered 1 hour before LPS stimulation [Sigma-Aldrich; L6529; 1 mg/ml stock in ddH₂O; final concentration in well, 100 ng/ml (dilutions in medium)]. Cells treated with vehicle, cells treated with LPS alone, and cells treated with LPS plus reference item (dexamethasone, 10 μ M; Sigma-Aldrich, D4902) served as controls. After 24 hours of stimulation, cell supernatants were collected for the cytokine measurement (V-PLEX Proinflammatory Panel 1 Mouse Kit, K15048D, Mesoscale) and cells were subjected to 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) assay.

A β clearance

For A β clearance assay, 20,000 BV2 cells per well (uncoated 96-well plates) were plated out. After changing cells to treatment medium, drug compounds were administered 1 hour before A β stimulation [Bachem, 4061966; final concentration in well, 200 ng/ml (dilutions in medium)]. Cells treated with vehicle and cells treated with A β alone served as controls. After 3 hours of A β stimulation, cell supernatants were collected for the A β measurement and cells were carefully washed twice with PBS and thereafter lysed in 35 μ l of cell lysis buffer [50 mM tris-HCl (pH 7.4), 150 mM NaCl, 5 mM EDTA, and 1% SDS] supplemented with protease inhibitors. Supernatants and cell lysates were analyzed for human A β 42 with the MSD V-PLEX Human A β 42 Peptide (6E10) Kit (K151LBE, Mesoscale Discovery). The immune assay was carried out according to the manual, and plates were read on MESO QuickPlex SQ 120.

Tau phosphorylation

SH-SY5Y-hTau441(V337M/R406W) cells were maintained in culture medium [DMEM medium, 10% FCS, 1% nonessential amino acids (NEAA), 1% L-glutamine, gentamycin (100 μ g/ml), and geneticin G-418 (300 μ g/ml)] and differentiated with 10 μ M retinoic acid for 5 days changing medium every 2 to 3 days. Before the treatment, cells were seeded onto 24-well plates at a cell density of 2×10^5 cells per well on day one of in vitro culture (DIV1). Drug compounds were applied on DIV2. After 24 hours of incubation (DIV3), cells on 24-well plates were harvested in 60 μ l of RIPA buffer [50 mM tris (pH 7.4), 1% NP-40, 0.25% Na-deoxycholate, 150 mM NaCl, 1 mM EDTA supplemented with freshly added 1 μ M NaF, 0.2 mM Na-orthovanadate, 80 μ M glycerophosphate, protease (Calbiochem), and phosphatase (Sigma-Aldrich) inhibitor cocktail]. Protein concentration was determined by BCA assay (Pierce, Thermo Fisher Scientific), and samples were adjusted to a uniform total protein concentration. Total tau and phosphorylated tau were determined by immunosorbent assay from Mesoscale Discovery [Phospho(Thr231)/Total Tau Kit K15121D, Mesoscale Discovery].

Trophic factor withdrawal

Primary cortical neurons from embryonic day 18 (E18) C57Bl/6 mice were prepared as previously described. On the day of preparation (DIV1), cortical neurons were seeded on poly-D-lysine

precoated 96-well plates at a density of 3×10^4 cells per well. Every 4 to 6 days, a half medium exchange using full medium (Neurobasal, 2% B-27, 0.5 mM glutamine, and 1% penicillin-streptomycin) was carried out. On DIV8, a full medium exchange to B-27 free medium (Neurobasal, 0.5 mM glutamine, and 1% penicillin-streptomycin) was performed and drug compounds were applied thereafter. The experiment was carried out with $n = 6$ technical replicates per condition, and vehicle-treated cells served as control. After 28 hours on B-27-free medium, cells were subjected to YO-PRO/propidium iodide (PI) and MTT as well as lactate dehydrogenase (LDH) assay.

MTT assay

MTT solution was added to each well in a final concentration of 0.5 mg/ml. After 2 hours, the MTT-containing medium was aspirated. Cells were lysed in 3% SDS, and the formazan crystals were dissolved in isopropanol/HCl. Optical density was measured with a Cytation 5 (BioTek) multimode reader at wavelength 570 nm. Values were calculated as percent of control values (vehicle control or lesion control).

LDH assay

Supernatants collected after treatment were subjected to the LDH toxicity assay by using the Cytotoxicity Detection Kit (Roche Diagnostics, catalog no. 11 644 793 001). Seventy microliters of cell culture supernatant was transferred to clear 96-well plates. Seventy microliters of freshly prepared reaction mixture was added to each well, and the mixture was incubated for 1 hour at RT protected from light. Absorbance was measured at 492 and 620 nm as reference wavelength with a Cytation 5 (BioTek) multimode reader. Values of culture medium were subtracted as background control. Values were calculated as percent of control values (vehicle control or lesion control).

YO-PRO/PI apoptosis and necrosis assay

YO-PRO-1 (Invitrogen; Y3603) assay was carried out to detect apoptotic cells in combination with PI (P4864 Sigma-Aldrich) staining for necrotic cells.

Part of supernatant of the cultivated cells was sucked off so that 90 μ l was remaining per well. YO-PRO-1 solution (50 μ M) was prepared out of the 1 mM YO-PRO-1 stock solution in dimethyl sulfoxide (DMSO). The stock solution was diluted in a ratio of 1:20 in PBS, and PI was added to the same stock to a final concentration of 1 μ g/ml. Ten microliters of this 50 μ M YO-PRO-1/PI (1 μ g/ml) solution in PBS was added to the remaining 90 μ l to result in a final concentration of 5 μ M YO-PRO-1 in well. Incubation for 15 min in the incubator at 37°C was performed (light protected). Supernatant was sucked off completely and discarded. PBS (140 μ l) was added to well. Plate was measured at the multimode reader (Cytation 5, BioTek).

A β secretion

H4-hAPP cells were cultivated in Opti-MEM supplemented with 10% FCS, 1% penicillin/streptomycin, hygromycin B (200 μ g/ml), and blasticidin S (2.5 μ g/ml).

H4-hAPP cells were seeded into 96-well plates (2×10^4 cells per well). On the next day, cells in 96-well plates were treated with compounds, reference item [400 nM N-[N-(3,5-difluorophenacetyl)-l-alanyl]-S-phenylglycine t-butyl ester (DAPT)], or vehicle. Twenty-four hours later, supernatants were collected for further A β measurements by MSD [V-PLEX A β Peptide Panel 1 (6E10) Kit, K15200E, Mesoscale Discovery].

Statistical analyses

Statistical analysis was performed in GraphPad Prism 9.1.2. Group differences were evaluated for each test item separately by one-way

analysis of variance (ANOVA) followed by Dunnett's multiple comparison test versus vehicle or lesion control.

SUPPLEMENTARY MATERIALS

Supplementary material for this article is available at <https://science.org/doi/10.1126/sciadv.abi8178>

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A brain proteomic signature of incipient Alzheimer's disease in young *APOE* #4 carriers identifies novel drug targets

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